Regression Analysis

Lecture notes, Chapters 3–16

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Chapter 3. Simple Linear Regression

3.1 Regression and the straight-line model

Regression studies how a **response** Y changes with **explanatory** variables. Galton observed that sons of unusually tall or short fathers are closer to the average: heights "regress" to the mean. In Pearson's father–son data the slope is about 0.5: a father 6 inches above average has sons about 3 inches above average.

With one explanatory variable and a linear relationship we speak of **simple linear regression**.

Definition (Simple linear regression model)

For data $(x_1, y_1), \ldots, (x_n, y_n)$,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2), \qquad i = 1, \ldots, n,$$

where the x_i are treated as fixed (nonrandom) constants and $\beta_0, \beta_1, \sigma^2$ are unknown.

The model packs three assumptions.

(1) **Linearity**: $E(Y \mid X = x) = \beta_0 + \beta_1 x$.

(2) **Normality and equal variance**: given $X = x$, $Y \sim N(\beta_0 + \beta_1 x, \sigma^2)$, with σ^2 not depending on x (**homoscedasticity**); otherwise the errors are **heteroscedastic** (Chapter 12).

(3) **Independence**: the errors are uncorrelated, $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$ (independent under normality).

Linear *in the parameters*: $\beta_0 + \beta_1 x + \beta_2 x^2$ is linear. For random X the analysis is conditional on the x_i: the **regression of Y on X**, not of X on Y.

It is often convenient to centre the explanatory variable:

$$y_i = \beta_0^* + \beta_1 (x_i - \bar{x}) + \epsilon_i, \qquad \beta_0^* = \beta_0 + \beta_1 \bar{x}. \tag{3.1}$$

3.2 Least squares

Motivation. Least squares minimizes squared vertical deviations, with closed-form estimates equal to the MLE under normal errors.

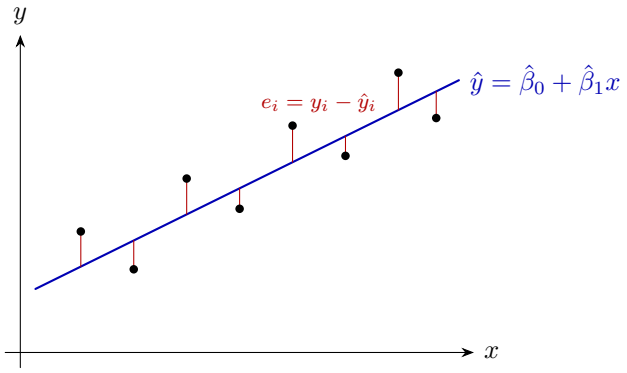


Figure. *Least squares chooses the line minimizing the sum of the squared vertical deviations e_i (red), not perpendicular distances.*

Definition (Least squares estimator)

The **least squares estimators** (LSE) $\hat\beta_0, \hat\beta_1$ minimize

$$S(\beta_0,\beta_1)=\sum_{i=1}^{n}(y_i-\beta_0-\beta_1 x_i)^2.$$

Write

$$S_{xx}=\sum(x_i-\bar x)^2,\qquad S_{yy}=\sum(y_i-\bar y)^2,\qquad S_{xy}=\sum(x_i-\bar x)(y_i-\bar y).$$

Theorem 3.1 (Normal equations)

If $S_{xx}>0$, the LSE is the unique solution of the **normal equations**

$$n\hat\beta_0+\hat\beta_1\sum x_i=\sum y_i,\qquad \hat\beta_0\sum x_i+\hat\beta_1\sum x_i^2=\sum x_iy_i,$$

namely

$$\hat\beta_1=\frac{S_{xy}}{S_{xx}},\qquad \hat\beta_0=\bar y-\hat\beta_1\bar x.$$

Proof. Setting the partial derivatives to zero,

$$\frac{\partial S}{\partial\beta_0}=-2\sum(y_i-\beta_0-\beta_1x_i)=0,\qquad \frac{\partial S}{\partial\beta_1}=-2\sum x_i(y_i-\beta_0-\beta_1x_i)=0,$$

gives the normal equations. The first gives $\hat\beta_0=\bar y-\hat\beta_1\bar x$; substituting into the second, $\hat\beta_1(\sum x_i^2-n\bar x^2)=\sum x_iy_i-n\bar x\bar y$, i.e. $\hat\beta_1S_{xx}=S_{xy}$. The Hessian

$$J=2\begin{pmatrix}n & \sum x_i\\ \sum x_i & \sum x_i^2\end{pmatrix}$$

has $n>0$ and determinant $4nS_{xx}>0$, so it is positive definite. S is a convex quadratic, and the stationary point is its unique minimum. ∎

In the centred form (3.1) the normal equations decouple: $\hat\beta_0^{*}=\bar y$ and $\hat\beta_1=S_{xy}/S_{xx}$, so the fitted line is

$$\hat y=\bar y+\hat\beta_1(x-\bar x).$$

Example 3.1

Advertising spending x and sales y for six stores:

x	2	3	5	6	8	10
y	5	7	8	11	13	16

Here $\bar x=34/6=5.667$, $\bar y=10$, $S_{xx}=238-34^2/6=45.33$ and $S_{xy}=401-34\cdot60/6=61$. So $\hat\beta_1=61/45.33=1.346$, $\hat\beta_0=10-1.346\cdot5.667=2.37$, and $\hat y=2.37+1.346x$.

Maximum likelihood. Under normal errors the log-likelihood is

$$l(\beta_0,\beta_1,\sigma^2)=-\frac{n}{2}\log(2\pi\sigma^2)-\frac{1}{2\sigma^2}S(\beta_0,\beta_1).$$

For each σ^2 it is maximized by minimizing S, so the MLEs of β_0,β_1 are the LSEs, and $\hat\sigma^2_{\mathrm{MLE}}=S(\hat\beta_0,\hat\beta_1)/n$.

3.3 Properties of the fitted line

The **fitted values** are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the **residuals** are $e_i = y_i - \hat{y}_i$.

Theorem 3.2

(a) $\sum e_i = 0$; equivalently $\sum y_i = \sum \hat{y}_i$.

(b) $\sum x_i e_i = 0$.

(c) $\sum \hat{y}_i e_i = 0$.

(d) The fitted line passes through $(\bar{x}, \bar{y})$.

(e) $\sum e_i^2$ is the minimum of $S(\beta_0, \beta_1)$.

Proof. (a) and (b) are the two normal equations, since $\sum e_i = \sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$ and $\sum x_i e_i = \sum x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$.

(c) $\sum \hat{y}_i e_i = \hat{\beta}_0 \sum e_i + \hat{\beta}_1 \sum x_i e_i = 0$.

(d) $\hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$.

(e) is the definition of the LSE. ■

Geometrically the residual vector is orthogonal to $\mathbf{1}$, $\mathbf{x}$ and the fitted vector, so e_i against $\hat{y}_i$ (or x_i) shows no linear trend.

3.4 Decomposition of variation

Motivation. SST splits into regression and residual parts; R^2 is their ratio.

Theorem 3.3 (Sum of squares decomposition)

$$\underbrace{\sum (y_i - \bar{y})^2}_{SST} = \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE} + \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR},$$

and $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$.

Proof. Write $y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$ and square. The cross term is $2\sum e_i(\hat{y}_i - \bar{y}) = 2\sum \hat{y}_i e_i - 2\bar{y}\sum e_i = 0$ by Theorem 3.2. Since $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$. ■

SST, SSE, SSR are the **total**, **residual** (error) and **regression** sums of squares, with $n-1$, $n-2$, 1 degrees of freedom.

Definition (Coefficient of determination)

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \in [0, 1].$$

R^2 is the proportion of variation explained: 1 for a perfect fit, 0 for $\hat{\beta}_1 = 0$. The standard error of estimate is $s = \sqrt{MSE}$, $MSE = SSE/(n-2)$, unbiased for σ^2 (Chapter 4).

Example 3.2

In Example 3.1, $\sum y_i^2 = 684$, so $SST = S_{yy} = 684 - 600 = 84$. Then $SSR = 61^2/45.33 = 82.08$, $SSE = 84 - 82.08 = 1.92$ and $R^2 = 0.977$: advertising explains about 98% of the variation in sales.

3.5 Correlation

For a random pair (X, Y) the **population correlation** is $\rho_{XY} = \mathrm{Cov}(X, Y)/\sqrt{\mathrm{Var}\, X\, \mathrm{Var}\, Y}$, and the **sample correlation** is

$$r_{xy} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \in [-1, 1].$$

Theorem 3.4

(a) $R^2 = r_{xy}^2$ in simple linear regression.

(b) $r_{xy} = \hat{\beta}_1\, s_x/s_y$, where $s_x^2 = S_{xx}/(n-1)$ and $s_y^2 = S_{yy}/(n-1)$; hence r_{xy} has the sign of $\hat{\beta}_1$.

Proof. (a) By Theorem 3.3, $R^2 = \frac{SSR}{SST} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r_{xy}^2$.

(b) $r_{xy} = \frac{S_{xy}}{S_{xx}} \sqrt{\frac{S_{xx}}{S_{yy}}} = \hat{\beta}_1 \frac{s_x}{s_y}$. ■

r_{xy} measures only *linear* association. For $x = 0, 1, 2, 3, 4$ and $y = x^2 - 4x + 6 = (6, 3, 2, 3, 6)$, $S_{xy} = 0$ and $r_{xy} = 0$, although y is a deterministic function of x.

Remark

$\hat{\beta}_1$ is the expected change in y per unit x; r_{xy} is unit-free and symmetric. The regression of x on y has slope S_{xy}/S_{yy}; the slopes multiply to $r_{xy}^2 \le 1$, so the lines coincide only if $|r_{xy}| = 1$.

3.6 Selected exercises

Motivation. Least squares on special designs: through the origin, two distinct x values, and the slope as an average of pairwise slopes.

Exercise 3.1 (Regression through the origin)

Consider $y_i = \beta x_i + \epsilon_i$, $i = 1, \ldots, n$, with $\sum x_i^2 > 0$ and no intercept.
(a) Show that the least squares estimator is $\hat{\beta} = \sum x_i y_i / \sum x_i^2$ and that the residuals $e_i = y_i - \hat{\beta} x_i$ satisfy $\sum x_i e_i = 0$, but in general $\sum e_i \neq 0$.
(b) Show that $\sum y_i^2 = \sum \hat{y}_i^2 + \sum e_i^2$, and give an example in which the centred decomposition $SST = SSR + SSE$ of Theorem 3.3 fails.

Solution [Supplement].

(a) $S(\beta) = \sum (y_i - \beta x_i)^2$ has $S'(\beta) = -2 \sum x_i (y_i - \beta x_i)$, $S'' = 2 \sum x_i^2 > 0$: strictly convex, minimized where $\sum x_i e_i = 0$. Nothing forces $\sum e_i = 0$: $x = (1, 2)$, $y = (1, 1)$ give $\hat{\beta} = 3/5$, $e = (0.4, -0.2)$, $\sum e_i = 0.2$.

(b) Since $\hat{y}_i = \hat{\beta} x_i$,

$$\sum y_i^2 = \sum (\hat{y}_i + e_i)^2 = \sum \hat{y}_i^2 + 2\hat{\beta} \sum x_i e_i + \sum e_i^2 = \sum \hat{y}_i^2 + \sum e_i^2.$$

There $\bar{y} = 1$, $SST = 0$, $SSE = 0.2 > 0$: $SST \neq SSR + SSE$ and R^2 is meaningless. Without intercept use the uncorrected $\sum y_i^2$, $\sum \hat{y}_i^2$. ■

Exercise 3.2 (Two design points)

Let the x_i take two values $a < b$, $n_a, n_b \geq 1$ times ($n = n_a + n_b$), with response means $\bar{y}_a$, $\bar{y}_b$. Show that

$$S_{xx} = \frac{n_a n_b (b-a)^2}{n}, \qquad \hat{\beta}_1 = \frac{\bar{y}_b - \bar{y}_a}{b-a},$$

and that the fitted line passes through $(a, \bar{y}_a)$ and $(b, \bar{y}_b)$.

Solution [Supplement].

Step 1. $\bar{x} = (n_a a + n_b b)/n$, so $a - \bar{x} = -n_b(b-a)/n$ and $b - \bar{x} = n_a(b-a)/n$.

Step 2. Hence

$$S_{xx} = n_a \frac{n_b^2 (b-a)^2}{n^2} + n_b \frac{n_a^2 (b-a)^2}{n^2} = \frac{n_a n_b (b-a)^2 (n_a + n_b)}{n^2} = \frac{n_a n_b (b-a)^2}{n}.$$

Step 3. Using $S_{xy} = \sum (x_i - \bar{x}) y_i$ and grouping the observations by their x value,

$$S_{xy} = -\frac{n_b (b-a)}{n} n_a \bar{y}_a + \frac{n_a (b-a)}{n} n_b \bar{y}_b = \frac{n_a n_b (b-a)(\bar{y}_b - \bar{y}_a)}{n},$$

and $\hat{\beta}_1 = S_{xy}/S_{xx} = (\bar{y}_b - \bar{y}_a)/(b-a)$.

Step 4. The fitted line is $\hat{y}(x) = \bar{y} + \hat{\beta}_1 (x - \bar{x})$ with $\bar{y} = (n_a \bar{y}_a + n_b \bar{y}_b)/n$. At $x = a$, by Step 1,

$$\hat{y}(a) = \frac{n_a \bar{y}_a + n_b \bar{y}_b}{n} - \frac{n_b (\bar{y}_b - \bar{y}_a)}{n} = \frac{(n_a + n_b) \bar{y}_a}{n} = \bar{y}_a,$$

and in the same way $\hat{y}(b) = \bar{y}_b$. So the line joins the two group means, and all the variation within the groups goes into SSE. ■

Exercise 3.3 (The slope as an average of pairwise slopes)

Assume $S_{xx} > 0$. For $i < j$ with $x_i \neq x_j$, let $s_{ij} = (y_j - y_i)/(x_j - x_i)$ be the slope of the line through observations i and j. Show that

$$\hat{\beta}_1 = \sum_{i<j,\ x_i \neq x_j} w_{ij} s_{ij}, \qquad w_{ij} = \frac{(x_j - x_i)^2}{\sum_{k<l} (x_l - x_k)^2},$$

a weighted average of the pairwise slopes with weights proportional to the squared distance between the x values.

Solution [Supplement].

Step 1. For any numbers u_i, v_i, expanding the double sum,

$$\sum_{i<j} (u_j - u_i)(v_j - v_i) = \frac{1}{2} \sum_{i,j} (u_j - u_i)(v_j - v_i) = n \sum_i u_i v_i - \Big(\sum_i u_i \Big) \Big(\sum_i v_i \Big) = n \sum_i (u_i - \bar{u})(v_i - \bar{v}).$$

Step 2. With $u = v = x$ this gives $\sum_{k<l}(x_l - x_k)^2 = nS_{xx} > 0$, so the weights are well defined, nonnegative and sum to 1 (pairs with $x_i = x_j$ have zero weight). With $u = x$, $v = y$ it gives $\sum_{i<j}(x_j - x_i)(y_j - y_i) = nS_{xy}$.

Step 3. For a pair with $x_i \neq x_j$, $(x_j - x_i)(y_j - y_i) = (x_j - x_i)^2 s_{ij}$, and pairs with $x_i = x_j$ contribute 0 to the sum in Step 2. Therefore

$$\sum_{i<j,\ x_i \neq x_j} w_{ij} s_{ij} = \frac{\sum_{i<j}(x_j - x_i)(y_j - y_i)}{nS_{xx}} = \frac{nS_{xy}}{nS_{xx}} = \hat{\beta}_1.$$

Pairs of points far apart in x determine their slope more reliably, and least squares gives them more weight. ■

Chapter 4. Inference in Simple Regression

Throughout, $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ and fixed x_i with $S_{xx} > 0$.

4.1 Sampling distributions of the estimators

Motivation. To judge a fitted slope we need its sampling law: normal, and t when studentized.

Both estimators are linear in y:

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \sum_i k_i y_i, \qquad k_i = \frac{x_i - \bar x}{S_{xx}}, \qquad \sum k_i = 0, \quad \sum k_i x_i = 1, \quad \sum k_i^2 = \frac{1}{S_{xx}},$$

(using $S_{xy} = \sum (x_i - \bar x) y_i$), and $\hat\beta_0 = \bar y - \hat\beta_1 \bar x$.

Theorem 4.1

(a) $\hat\beta_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$.

(c) $\hat\beta_0 \sim N\Big(\beta_0,\ \sigma^2\big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\big)\Big)$ and $\operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\dfrac{\bar x \sigma^2}{S_{xx}}$.

Proof. (a) $\hat\beta_1$ is a linear combination of independent normals, with mean $\sum k_i(\beta_0 + \beta_1 x_i) = \beta_1$ and variance $\sigma^2 \sum k_i^2 = \sigma^2/S_{xx}$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = \sum_i \frac{1}{n} k_i \sigma^2 = 0$.

(c) $E\hat\beta_0 = (\beta_0 + \beta_1 \bar x) - \bar x \beta_1 = \beta_0$ and, by (b),

$$\operatorname{Var} \hat\beta_0 = \operatorname{Var} \bar y + \bar x^2 \operatorname{Var} \hat\beta_1 = \sigma^2\big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\big), \qquad \operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\bar x \operatorname{Var} \hat\beta_1.$$

■

Theorem 4.2

Let $SSE = \sum e_i^2$ and $MSE = SSE/(n-2)$. Then $SSE/\sigma^2 \sim \chi^2(n-2)$, independent of $(\hat\beta_0, \hat\beta_1)$. In particular $E(MSE) = \sigma^2$.

(The case $p = 1$ of Theorem 6.4; directly, use an orthogonal transformation whose first two rows span $\mathbf{1}$ and $\mathbf{x} - \bar x \mathbf{1}$.) Consequently

$$\frac{\hat\beta_1 - \beta_1}{\sqrt{MSE/S_{xx}}} \sim t(n-2), \qquad \frac{\hat\beta_0 - \beta_0}{\sqrt{MSE(1/n + \bar x^2/S_{xx})}} \sim t(n-2).$$

Corollary 4.3 (Intervals and tests for the coefficients)

$100(1-\alpha)\%$ confidence intervals are

$$\beta_1 :\ \hat\beta_1 \pm t_{\alpha/2}(n-2)\sqrt{\frac{MSE}{S_{xx}}}, \qquad \beta_0 :\ \hat\beta_0 \pm t_{\alpha/2}(n-2)\sqrt{MSE\big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\big)}.$$

The test of $H_0 : \beta_1 = 0$ rejects when $|t_0| > t_{\alpha/2}(n-2)$, $t_0 = \hat\beta_1/\sqrt{MSE/S_{xx}}$. If σ^2 is known, replace MSE by σ^2 and t by z.

Theorem 4.4 (ANOVA F test)

$E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(SSE) = (n-2)\sigma^2$. The statistic $F_0 = MSR/MSE$, $MSR = SSR/1$, satisfies $F_0 = t_0^2$, so the F test with $F_\alpha(1, n-2) = t_{\alpha/2}(n-2)^2$ is the two-sided t test of $\beta_1 = 0$.

Proof. $E(SSR) = S_{xx}E\hat\beta_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Also $SSR/MSE = \hat\beta_1^2 S_{xx}/MSE = t_0^2$. ■

So $E(MSR)/E(MSE) = 1 + \beta_1^2 S_{xx}/\sigma^2$, which is 1 exactly when $\beta_1 = 0$; large F_0 is evidence against H_0.

4.2 Estimating the mean response and predicting

Motivation. Predicting a new response must include its own error, so prediction intervals are wider than mean-response intervals.

At a point x, the mean response $\mu_{y\cdot x} = \beta_0 + \beta_1 x$ is estimated by $\hat y = \hat\beta_0 + \hat\beta_1 x = \bar y + \hat\beta_1(x - \bar x)$.

Theorem 4.5

(a) $\hat y \sim N\Big(\mu_{y\cdot x},\ \sigma^2\Big(\dfrac{1}{n} + \dfrac{(x-\bar x)^2}{S_{xx}}\Big)\Big)$, and a $100(1-\alpha)\%$ confidence interval for $\mu_{y\cdot x}$ is

$$\hat y \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}\Big)}.$$

(b) For a new observation $y_s = \beta_0 + \beta_1 x + \epsilon_s$, independent of the sample, $y_s - \hat y \sim N(0, \sigma^2(1 + \frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}))$, and a $100(1-\alpha)\%$ prediction interval is

$$\hat y \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(1 + \frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}\Big)}.$$

Proof. (a) By Theorem 4.1(b), $\operatorname{Var}\hat y = \operatorname{Var}\bar y + (x-\bar x)^2 \operatorname{Var}\hat\beta_1$. Studentize with MSE, which is independent of $\hat y$.

(b) y_s is independent of $\hat y$, so the variances add. ■

Both are narrowest at $x = \bar x$ and widen quadratically (extrapolation is costly); the prediction interval keeps the irreducible σ^2 as $n \to \infty$.

Example 4.1

For the advertising data of Example 3.1 ($n = 6$, $\bar x = 5.667$, $S_{xx} = 45.33$, $MSE = 1.92/4 = 0.48$, $\hat y = 2.37 + 1.346x$), at $x = 7$: $\hat y = 11.79$, $\frac{1}{6} + \frac{(1.333)^2}{45.33} = 0.206$, and $t_{0.025}(4) = 2.776$. The 95% confidence interval for the mean is $11.79 \pm 2.776\sqrt{0.48 \cdot 0.206} = 11.79 \pm 0.87$, and the prediction interval is $11.79 \pm 2.776\sqrt{0.48 \cdot 1.206} = 11.79 \pm 2.11$.

4.3 Inference on the correlation

When (X, Y) is bivariate normal with correlation ρ, the conditional mean is linear with slope $\beta_1 = \rho\sigma_Y/\sigma_X$, so $\rho = 0 \iff \beta_1 = 0$.

Theorem 4.6

(a) The t statistic for $\beta_1 = 0$ equals $t_0 = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$, which is $t(n-2)$ under $\rho = 0$.

(b) (Fisher) $Z = \frac{1}{2}\log\frac{1+r}{1-r}$ is approximately $N\big(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\big)$ for moderate n (say $n \geq 25$).

Proof. (a) $SSE = SST(1-r^2)$ and $\hat\beta_1^2 S_{xx} = SSR = r^2 SST$, so

$$t_0^2 = \frac{\hat\beta_1^2 S_{xx}}{SSE/(n-2)} = \frac{(n-2)r^2}{1-r^2}.$$

The null distribution is conditionally $t(n-2)$ given the x_i, hence unconditionally as well.

(b) is the variance-stabilizing transformation of the asymptotic law $\sqrt{n}(r-\rho) \to N(0, (1-\rho^2)^2)$; the correction $n-3$ improves the approximation. ■

Part (b) gives tests of $H_0 : \rho = \rho_0 \neq 0$ and confidence intervals: $\tanh\big(Z \pm z_{\alpha/2}/\sqrt{n-3}\big)$ covers ρ with probability about $1-\alpha$.

4.4 Residuals

Motivation. Residuals reveal violations of linearity, constant variance and normality.

Residuals are our window on the errors. In vector form $\mathbf{e} = (I-H)\mathbf{y}$ with H the hat matrix (Chapter 6), and:

Theorem 4.7

$Ee_i = 0$ and

$$\operatorname{Var}(e_i) = \sigma^2\Big(1 - \frac{1}{n} - \frac{(x_i-\bar x)^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(e_i, e_j) = -\sigma^2\Big(\frac{1}{n} + \frac{(x_i-\bar x)(x_j-\bar x)}{S_{xx}}\Big) \ (i \neq j).$$

Proof. $e_i = y_i - \bar y - \hat\beta_1(x_i - \bar x)$. Using $\operatorname{Var}\bar y = \sigma^2/n$, $\operatorname{Var}\hat\beta_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(y_i, \bar y) = \sigma^2/n$, $\operatorname{Cov}(y_i, \hat\beta_1) = \sigma^2 k_i$ and $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$,

$$\operatorname{Var} e_i = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}} - \frac{2}{n} - 2\frac{(x_i-\bar x)^2}{S_{xx}}\Big),$$

which simplifies to the claim; the covariance is computed in the same way. ■

So residuals are dependent with unequal variances (smaller far from $\bar x$; Chapter 10). As $\sum e_i = \sum x_i e_i = 0$, plots against x_i or $\hat y_i$ have no linear trend by construction; look for *non-random* structure:

- a curved band suggests a missing quadratic term or a transformation;
- a funnel suggests nonconstant variance (weighted least squares or a variance-stabilizing transformation of y);
- a trend in time order suggests correlated errors (Chapter 12).

4.5 Selected exercises

Motivation. Independence of the coefficient estimators, the size of the one-sided t test, $E(SSE)$ from residual variances, and design control of $\operatorname{Var}\hat\beta_1$.

Exercise 4.1 (Independence of the coefficient estimators)

Show that $\bar{y}$ and $\hat{\beta}_1$ are always independent, and that $\hat{\beta}_0$ and $\hat{\beta}_1$ are independent if and only if $\bar{x} = 0$.

Solution.

Step 1. (Joint normality.) Each of $\bar{y}$, $\hat{\beta}_0$, $\hat{\beta}_1$ is a fixed linear combination of $\mathbf{y} = (y_1, \dots, y_n)'$, so $(\hat{\beta}_0, \hat{\beta}_1, \bar{y})' = A\mathbf{y}$ for a fixed $3 \times n$ matrix A. Since $\mathbf{y} \sim N_n(X\beta, \sigma^2 I)$, $A\mathbf{y}$ is multivariate normal (Section 5.4).

Step 2. (Uncorrelated implies independent.) For jointly normal (U, W) with Cov $= 0$ the mgf $\exp\{t_1 EU + t_2 EW + \frac{1}{2}(t_1^2 \operatorname{Var} U + t_2^2 \operatorname{Var} W)\}$ factors; conversely independence gives zero covariance.

Step 3. By Theorem 4.1(b), $\operatorname{Cov}(\bar{y}, \hat{\beta}_1) = 0$, so $\bar{y}$ and $\hat{\beta}_1$ are independent for every design.

Step 4. $\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$ (Theorem 4.1(c)) vanishes iff $\bar{x} = 0$, so by Step 2 independence holds iff $\bar{x} = 0$ (then $\hat{\beta}_0 = \bar{y}$). ■

Exercise 4.2 (A one-sided test with a composite null)

Let $0 < \alpha < 1$ and test $H_0 : \beta_1 \le 0$ against $H_1 : \beta_1 > 0$. Show that the test rejecting H_0 when

$$t_0 = \frac{\hat{\beta}_1}{\sqrt{MSE/S_{xx}}} > t_\alpha(n-2)$$

satisfies

$$\sup_{\beta_0 \in \mathbb{R},\ \beta_1 \le 0,\ \sigma^2 > 0} P_{\beta_0, \beta_1, \sigma^2}(\text{the test rejects } H_0) = \alpha,$$

and that its rejection probability exceeds α whenever $\beta_1 > 0$.

Solution.

Step 1. For every parameter value, $T = (\hat{\beta}_1 - \beta_1)/\sqrt{MSE/S_{xx}} \sim t(n-2)$ (Section 4.1). Since $MSE > 0$ with probability one,

$$t_0 = T + \frac{\beta_1}{\sqrt{MSE/S_{xx}}}.$$

Step 2. If $\beta_1 \le 0$, the last term is ≤ 0, so $t_0 \le T$ and $\{t_0 > t_\alpha(n-2)\} \subseteq \{T > t_\alpha(n-2)\}$. Hence the rejection probability is at most $P(T > t_\alpha(n-2)) = \alpha$.

Step 3. At $\beta_1 = 0$, $t_0 = T$ and the rejection probability is exactly α for all β_0, σ^2; with Step 2 the supremum over $\{\beta_1 \le 0\}$ is α, attained at $\beta_1 = 0$.

Step 4. (Power.) $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$ is independent of MSE (Theorem 4.2). Write $c = t_\alpha(n-2)$ and $\tau = \sigma/\sqrt{S_{xx}}$. Conditioning on MSE,

$$P(t_0 > c) = E\Big[P\big(\hat{\beta}_1 > c\sqrt{MSE/S_{xx}} \mid MSE\big)\Big] = E\Big[1 - \Phi\Big(\frac{c\sqrt{MSE/S_{xx}} - \beta_1}{\tau}\Big)\Big].$$

The integrand increases strictly in β_1, so the rejection probability does too: it exceeds α for $\beta_1 > 0$ (unbiased test; this also reproves Step 2). ■

Exercise 4.3 (Expected residual sum of squares without chi-square theory)

(a) Using only Theorem 4.7, show that $E(SSE) = (n-2)\sigma^2$.
(b) Show that $\operatorname{Var} \hat{y}_i + \operatorname{Var} e_i = \sigma^2$ for each i, where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, and that $\sum_i \operatorname{Var} \hat{y}_i = 2\sigma^2$.

Solution [Supplement].

(a) Since $Ee_i = 0$, $Ee_i^2 = \operatorname{Var} e_i$. Summing the formula of Theorem 4.7 and using $\sum(x_i - \bar{x})^2 = S_{xx}$,

$$E(SSE) = \sum_i Ee_i^2 = \sigma^2\Big(n - 1 - \frac{S_{xx}}{S_{xx}}\Big) = (n-2)\sigma^2.$$

No normality is needed: only $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I$ were used.

(b) By Theorem 4.5(a) at $x = x_i$, $\operatorname{Var}\hat{y}_i = \sigma^2\{1/n + (x_i - \bar{x})^2/S_{xx}\}$; adding $\operatorname{Var} e_i$ from Theorem 4.7 gives σ^2. Summing over i,

$$\sum_i \operatorname{Var}\hat{y}_i = \sigma^2\Big(1 + \frac{S_{xx}}{S_{xx}}\Big) = 2\sigma^2.$$

So the total variance $n\sigma^2$ of the y_i splits into $2\sigma^2$ in the fitted values (two parameters) and $(n-2)\sigma^2$ in the residuals. ■

Exercise 4.4 (The best design for the slope)

Let n be even, and suppose the experimenter may choose each x_i in a fixed interval $[a, b]$. Show that

$$\operatorname{Var}\hat{\beta}_1 \geq \frac{4\sigma^2}{n(b-a)^2},$$

with equality if and only if half of the x_i equal a and the other half equal b.

Solution [Supplement].

Step 1. For any constant c, $\sum(x_i - c)^2 = S_{xx} + n(\bar{x} - c)^2 \geq S_{xx}$ (expand $x_i - c = (x_i - \bar{x}) + (\bar{x} - c)$; the cross term vanishes). Take $c = m = (a+b)/2$.

Step 2. Since $x_i \in [a, b]$, $|x_i - m| \leq (b-a)/2$. Hence

$$S_{xx} \leq \sum(x_i - m)^2 \leq \frac{n(b-a)^2}{4}, \qquad \operatorname{Var}\hat{\beta}_1 = \frac{\sigma^2}{S_{xx}} \geq \frac{4\sigma^2}{n(b-a)^2}.$$

Step 3. Equality in Step 2 iff $\bar{x} = m$ and $|x_i - m| = (b-a)/2$, i.e. all $x_i \in \{a, b\}$ with $n_a a + (n - n_a)b = n(a+b)/2$, i.e. $n_a = n/2$; this design gives $S_{xx} = n(b-a)^2/4$ (Exercise 3.2).

It cannot reveal curvature; in practice some interior points check linearity. ■

Chapter 5. Transformations, Comparisons and Quadratic Forms

5.1 Transformations and the Box–Cox family

Curvature or a funnel in the scatter or residual plot calls for transforming y (or x), e.g. $\log y$, $\sqrt{y}$, $1/y$, to restore linearity, constant variance and normality. Box and Cox embedded them in one family:

$$y^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log y, & \lambda = 0, \end{cases} \qquad y > 0,$$

which is continuous in λ since $(y^\lambda - 1)/\lambda \to \log y$. $\lambda = 1$ is (essentially) no transformation, $\lambda = \frac{1}{2}$ the square root and $\lambda = -1$ the reciprocal. The model is

$$y_i^{(\lambda)} = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

with λ an additional parameter.

Choosing λ. Comparing SSE across λ directly is meaningless, because the scale of $y^{(\lambda)}$ changes with λ. Divide by the Jacobian: with the geometric mean $\dot{y} = (\prod y_i)^{1/n}$, set

$$z_i^{(\lambda)} = \begin{cases} \dfrac{y_i^\lambda - 1}{\lambda \dot{y}^{\lambda-1}}, & \lambda \neq 0, \\ \dot{y} \log y_i, & \lambda = 0. \end{cases}$$

Then fit $z^{(\lambda)} = \beta_0 + \beta_1 x + \epsilon$ for a grid of λ and choose the λ with the smallest $SSE(\lambda)$.

Proposition 5.1

The profile log-likelihood of λ is $-\frac{n}{2} \log SSE(\lambda) + \text{const}$, $SSE(\lambda)$ the residual sum of squares of $z^{(\lambda)}$; so minimizing $SSE(\lambda)$ is maximum likelihood.

Proof. The density of y_i is the normal density of $y_i^{(\lambda)}$ times the Jacobian $y_i^{\lambda-1}$. Maximizing over β and σ^2 for fixed λ gives

$$l_{\max}(\lambda) = -\frac{n}{2} \log \frac{SSE_y(\lambda)}{n} - \frac{n}{2} + (\lambda - 1) \sum \log y_i,$$

where $SSE_y(\lambda)$ is the residual sum of squares for $y^{(\lambda)}$. Since $z^{(\lambda)} = y^{(\lambda)}/\dot{y}^{\lambda-1}$, $SSE(\lambda) = SSE_y(\lambda)\dot{y}^{-2(\lambda-1)}$, and $(\lambda - 1)\sum \log y_i = n(\lambda - 1)\log \dot{y}$. Substituting, $l_{\max}(\lambda) = -\frac{n}{2} \log SSE(\lambda) + \text{const}$. ∎

An approximate 95% confidence set for λ is $\{\lambda : 2\{l_{\max}(\hat{\lambda}) - l_{\max}(\lambda)\} \leq \chi^2_{0.05}(1)\}$; in practice one picks a simple value (0, $\frac{1}{2}$, 1) inside it.

5.2 Measurement error in the explanatory variable

True model $y_i = \beta_0 + \beta_1 x_i^* + \epsilon_i$, observed $x_i = x_i^* + \delta_i$ with δ_i (mean 0, variance σ_δ^2) independent of x_i^* and ϵ_i, and x_i^* random with variance σ_*^2. Then $y_i = \beta_0 + \beta_1 x_i + (\epsilon_i - \beta_1 \delta_i)$, with error correlated with x_i.

Proposition 5.2 (Attenuation)

The least squares slope from regressing y on the observed x satisfies

$$\hat{\beta}_1 \xrightarrow{p} \frac{\operatorname{Cov}(x,y)}{\operatorname{Var}(x)} = \beta_1 \frac{\sigma_*^2}{\sigma_*^2+\sigma_\delta^2} = \beta_1\Big(1-\frac{\sigma_\delta^2}{\operatorname{Var} x}\Big).$$

Proof. $\hat{\beta}_1 = \frac{\frac{1}{n}\sum(x_i-\bar{x})(y_i-\bar{y})}{\frac{1}{n}\sum(x_i-\bar{x})^2}$, a ratio of sample moments that converges to $\operatorname{Cov}(x,y)/\operatorname{Var}(x)$. Here $\operatorname{Cov}(x,y) = \operatorname{Cov}(x^*+\delta, \beta_1 x^* + \epsilon) = \beta_1\sigma_*^2$ and $\operatorname{Var} x = \sigma_*^2 + \sigma_\delta^2$. ■

The slope is biased towards zero (**attenuation**, regression dilution), little when σ_δ^2 is small relative to the spread of the x's.

5.3 Comparing two regression lines

Two groups, $i = 1, 2$, give

$$y_{ij} = \beta_{0i} + \beta_{1i}x_{ij} + \epsilon_{ij}, \qquad \epsilon_{ij} \overset{\text{iid}}{\sim} N(0,\sigma^2), \qquad j = 1,\dots,n_i,$$

with a *common* error variance. This is the **full model**.

Are the lines identical? Test $H_0 : \beta_{01} = \beta_{02},\ \beta_{11} = \beta_{12}$. Under H_0 we have the **reduced model** $y_{ij} = \beta_0 + \beta_1 x_{ij} + \epsilon_{ij}$ for the pooled $n_1 + n_2$ observations.

(1) Fit each group separately: $SSE(F) = SSE_1 + SSE_2$, with $df_F = n_1 + n_2 - 4$.

(2) Fit one line to the pooled data: $SSE(R)$, with $df_R = n_1 + n_2 - 2$.

(3) Compute

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/(df_R - df_F)}{SSE(F)/df_F} = \frac{\{SSE(R) - SSE(F)\}/2}{SSE(F)/(n_1+n_2-4)}.$$

(4) Reject H_0 when $F_0 > F_\alpha(2, n_1 + n_2 - 4)$.

That $F_0 \sim F(2, n_1 + n_2 - 4)$ under H_0 follows from the general theory of Chapters 7 and 8: $SSE(F)/\sigma^2 \sim \chi^2(df_F)$, and $\{SSE(R) - SSE(F)\}/\sigma^2 \sim \chi^2(2)$ independently (a general linear hypothesis with two constraints).

Are the lines parallel? For $H_0 : \beta_{11} = \beta_{12}$, the separate slope estimates are independent, so

$$\operatorname{Var}(\hat{\beta}_{11} - \hat{\beta}_{12}) = \sigma^2\Big(\frac{1}{S_{xx,1}} + \frac{1}{S_{xx,2}}\Big), \qquad t_0 = \frac{\hat{\beta}_{11} - \hat{\beta}_{12}}{\sqrt{MSE(F)\big(\frac{1}{S_{xx,1}} + \frac{1}{S_{xx,2}}\big)}} \sim t(n_1 + n_2 - 4)$$

under H_0, with $MSE(F) = SSE(F)/(n_1+n_2-4)$. Reject when $|t_0| > t_{\alpha/2}(n_1+n_2-4)$. The same comparisons can be done in a single regression with indicator variables (Chapter 13).

5.4 Multivariate normal distribution

The distribution theory of regression rests on the multivariate normal distribution and quadratic forms.

Definition (Multivariate normal)

$\mathbf{X} \sim N_n(\mu, \Sigma)$, with Σ symmetric nonnegative definite (possibly singular), if any of the following equivalent conditions holds:

(1) $\mathbf{X} \overset{d}{=} A\mathbf{Z} + \mu$ for some $n \times k$ matrix A with $AA' = \Sigma$ and $\mathbf{Z} \sim N_k(\mathbf{0}, I)$;

(2) $\mathbf{X} \overset{d}{=} \Sigma^{1/2}\mathbf{Z} + \mu$ with $\mathbf{Z} \sim N_n(\mathbf{0}, I)$;

(3) $Ee^{\mathbf{t}'\mathbf{X}} = \exp(\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t})$ for all $\mathbf{t} \in \mathbb{R}^n$;

(4) $\mathbf{a}'\mathbf{X} \sim N(\mathbf{a}'\mu, \mathbf{a}'\Sigma\mathbf{a})$ for every $\mathbf{a}$.

The equivalences follow from $Ee^{\mathbf{s}'\mathbf{Z}} = e^{\mathbf{s}'\mathbf{s}/2}$: for instance, (1) gives $Ee^{\mathbf{t}'(A\mathbf{Z}+\mu)} = e^{\mathbf{t}'\mu}e^{\frac{1}{2}\mathbf{t}'AA'\mathbf{t}}$, which is (3). Linear maps preserve normality: $B\mathbf{X} + \mathbf{c} \sim N(B\mu + \mathbf{c}, B\Sigma B')$, and jointly normal vectors are independent iff they are uncorrelated.

5.5 Quadratic forms

Noncentral chi-square: for independent $X_i \sim N(\mu_i, 1)$, $\sum_{i=1}^r X_i^2 \sim \chi^2(r, \lambda)$, $\lambda = \frac{1}{2}\sum \mu_i^2$ (the lecture's half convention; some books use $\sum \mu_i^2$).

Theorem 5.3

If $\mathbf{Z} \sim N(\mathbf{0}, I)$ and A is symmetric with $A^2 = A$, then $\mathbf{Z}'A\mathbf{Z} \sim \chi^2(r)$ with $r = \mathrm{tr}A$. Conversely, if $\mathbf{Z}'A\mathbf{Z}$ is chi-square, then $A^2 = A$.

Proof. $A = P\,\mathrm{diag}(\lambda_i)P'$, P orthogonal; $A^2 = A$ forces $\lambda_i \in \{0, 1\}$, with $\mathrm{tr}A = r$ ones. With $\mathbf{W} = P'\mathbf{Z} \sim N(\mathbf{0}, I)$, $\mathbf{Z}'A\mathbf{Z} = \sum_i \lambda_i W_i^2$. Converse: equality of the mgfs $\prod_i (1 - 2\lambda_i t)^{-1/2} = (1 - 2t)^{-r/2}$ forces $\lambda_i \in \{0, 1\}$. ■

Theorem 5.4 (Moments of quadratic forms)

Let $\mathbf{y} \sim N(\mu, V)$ and A symmetric. Then

(a) $E(\mathbf{y}'A\mathbf{y}) = \mathrm{tr}(AV) + \mu'A\mu$ (this needs only the mean and covariance);

(b) $\mathrm{Cov}(\mathbf{y}, \mathbf{y}'A\mathbf{y}) = 2VA\mu$;

(c) $\mathrm{Var}(\mathbf{y}'A\mathbf{y}) = 2\mathrm{tr}\{(AV)^2\} + 4\mu'AVA\mu$.

Proof. (a) $\mathbf{y}'A\mathbf{y} = \mathrm{tr}(A\mathbf{y}\mathbf{y}')$ and $E\mathbf{y}\mathbf{y}' = V + \mu\mu'$. (b), (c) Write $\mathbf{y} = \mu + \mathbf{w}$ with $\mathbf{w} \sim N(\mathbf{0}, V)$: $\mathbf{y}'A\mathbf{y} = \mathbf{w}'A\mathbf{w} + 2\mu'A\mathbf{w} + \mu'A\mu$. Odd moments of $\mathbf{w}$ vanish, so $\mathrm{Cov}(\mathbf{w}, \mathbf{w}'A\mathbf{w}) = 0$ and $\mathrm{Cov}(\mathbf{y}, \mathbf{y}'A\mathbf{y}) = \mathrm{Cov}(\mathbf{w}, 2\mu'A\mathbf{w}) = 2VA\mu$. Likewise $\mathrm{Var}(\mathbf{y}'A\mathbf{y}) = \mathrm{Var}(\mathbf{w}'A\mathbf{w}) + 4\mu'AVA\mu$. Write $\mathbf{w} = V^{1/2}\mathbf{z}$ and diagonalize $V^{1/2}AV^{1/2}$ with eigenvalues λ_i: then $\mathrm{Var}(\mathbf{w}'A\mathbf{w}) = 2\sum \lambda_i^2 = 2\mathrm{tr}\{(AV)^2\}$. ■

Theorem 5.5

Let $\mathbf{y} \sim N(\mu, V)$ with V nonsingular and A symmetric.

(a) $\mathbf{y}'A\mathbf{y} \sim \chi^2(r(A), \frac{1}{2}\mu'A\mu)$ if and only if AV is idempotent.

(b) In particular, if $\mathbf{y} \sim N(\mu, \sigma^2 I)$, then $\mathbf{y}'A\mathbf{y}/\sigma^2 \sim \chi^2(r(A), \frac{1}{2\sigma^2}\mu'A\mu)$ iff A is idempotent; and $\mathbf{y}'\mathbf{y}/\sigma^2 \sim \chi^2(n, \mu'\mu/(2\sigma^2))$.

Proof of sufficiency in (a). With $\mathbf{z} = V^{-1/2}\mathbf{y} \sim N(V^{-1/2}\mu, I)$, $B = V^{1/2}AV^{1/2}$: $\mathbf{y}'A\mathbf{y} = \mathbf{z}'B\mathbf{z}$, and $B^2 = V^{1/2}AVAV^{1/2} = B$ as $AVA = A$ (from $(AV)^2 = AV$). With $B = P\,\mathrm{diag}(I_r, 0)P'$ and $\mathbf{w} = P'\mathbf{z}$, $\mathbf{z}'B\mathbf{z} = \sum_{i\le r} w_i^2$, noncentral chi-square with $r = \mathrm{tr}B = r(A)$ and noncentrality $\frac{1}{2}\mu'V^{-1/2}BV^{-1/2}\mu = \frac{1}{2}\mu'A\mu$. Necessity: compare mgfs as in Theorem 5.3. ■

Theorem 5.6 (Independence)

Let $\mathbf{y} \sim N(\mu, V)$.

(a) $\mathbf{y}'A\mathbf{y}$ and $B\mathbf{y}$ are independent if $BVA = 0$.

(b) $\mathbf{y}'A\mathbf{y}$ and $\mathbf{y}'B\mathbf{y}$ (with A, B symmetric) are independent if $AVB = 0$.

(The conditions are also necessary.)

Proof. We give the proof when A and B are nonnegative definite (for instance idempotent), the case used in regression.

(a) $A = LL'$, L of full column rank; $\mathrm{Cov}(B\mathbf{y}, L'\mathbf{y}) = BVL = BVLL'L(L'L)^{-1} = BVAL(L'L)^{-1} = 0$, so $L'\mathbf{y} \perp B\mathbf{y}$, and $\mathbf{y}'A\mathbf{y} = \|L'\mathbf{y}\|^2$.

(b) The same argument applied to both forms: $\mathrm{Cov}(L_A'\mathbf{y}, L_B'\mathbf{y}) = L_A'VL_B = 0$ when $AVB = 0$. ■

Theorem 5.7 (Cochran)

Let $\mathbf{y} \sim N(\mathbf{0}, I_n)$ and let $A_1, \ldots, A_p$ be symmetric with $\sum A_j = I_n$ and $r(A_j) = k_j$. Then the forms $\mathbf{y}'A_j\mathbf{y}$ are independent with $\mathbf{y}'A_j\mathbf{y} \sim \chi^2(k_j)$ if and only if $\sum k_j = n$.

Proof of sufficiency. $\sum A_j = I$ and $\sum r(A_j) = n$ give $\mathbb{R}^n = \bigoplus_j \mathrm{col}(A_j)$. For $\mathbf{v} \in \mathrm{col}(A_k)$, $\mathbf{v} = \sum_j A_j\mathbf{v}$ and uniqueness gives $A_k\mathbf{v} = \mathbf{v}$, $A_j\mathbf{v} = \mathbf{0}$ $(j \neq k)$: $A_k^2 = A_k$, $A_jA_k = 0$. Apply Theorems 5.3 and 5.6. ■

5.6 Expected mean squares in simple regression

Motivation. Both mean squares estimate σ^2 when $\beta_1 = 0$; otherwise MSR is inflated by $\beta_1^2 S_{xx}$.

In the centred model $y_i = \beta_0^* + \beta_1(x_i - \bar{x}) + \epsilon_i$:

Proposition 5.8

$$E(SST) = \beta_1^2 S_{xx} + (n-1)\sigma^2, \qquad E(SSR) = \sigma^2 + \beta_1^2 S_{xx}, \qquad E(SSE) = (n-2)\sigma^2.$$

Proof. $y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (\epsilon_i - \bar{\epsilon})$, so $E(SST) = \beta_1^2 S_{xx} + E\sum(\epsilon_i - \bar{\epsilon})^2 = \beta_1^2 S_{xx} + (n-1)\sigma^2$ (the cross term has mean 0). $E(SSR) = S_{xx}E\hat{\beta}_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Subtract. ■

In matrix form, with $J_n = \mathbf{1}\mathbf{1}'$, $SST = \mathbf{y}'(I - J_n/n)\mathbf{y}$; $I - J_n/n$ is idempotent of rank $n-1$, and Theorem 5.4(a) reproduces $E(SST)$.

5.7 Selected exercises

Motivation. Exact distributions and variances from the quadratic-form theorems.

Exercise 5.1 (Quadratic forms in a projection)

Let $\mathbf{y} \sim N_n(\mathbf{0}, I_n)$, let X be a fixed $n \times k$ matrix of rank $k < n$, and $H = X(X'X)^{-1}X'$.
(a) Find the distribution of $\mathbf{y}'H\mathbf{y}$.
(b) Find the distribution of $\mathbf{y}'(I-H)\mathbf{y}$.
(c) Find the distribution of $T = \mathbf{y}'H\mathbf{y}/\mathbf{y}'(I-H)\mathbf{y}$.

Solution.

(a) H is symmetric, and $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$. Its rank equals its trace, $\mathrm{tr}\{(X'X)^{-1}X'X\} = \mathrm{tr}I_k = k$. By Theorem 5.3, $\mathbf{y}'H\mathbf{y} \sim \chi^2(k)$.

(b) $I-H$ is symmetric with $(I-H)^2 = I - 2H + H^2 = I - H$ and $\mathrm{tr}(I-H) = n-k$. By Theorem 5.3, $\mathbf{y}'(I-H)\mathbf{y} \sim \chi^2(n-k)$.

(c) $H(I-H) = 0$, so the forms are independent (Theorem 5.6(b), $V = I$); the $\chi^2(n-k)$ denominator is a.s. positive, and

$$\frac{n-k}{k}\,T = \frac{\mathbf{y}'H\mathbf{y}/k}{\mathbf{y}'(I-H)\mathbf{y}/(n-k)} \sim F(k, n-k).$$

Thus T is distributed as $\frac{k}{n-k}F$ with $F \sim F(k, n-k)$. ■

Exercise 5.2 (Variance of the residual mean square)

(a) Let $y_1, \ldots, y_n$ be i.i.d. $N(\mu, \sigma^2)$ and $S^2 = \sum(y_i - \bar{y})^2/(n-1)$. Use Theorem 5.4 to show $\mathrm{Var}\, S^2 = 2\sigma^4/(n-1)$.
(b) In the regression model $\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$ with X of size $n \times (p+1)$ and full column rank, show $\mathrm{Var}(MSE) = 2\sigma^4/(n-p-1)$.

Solution [Supplement].

(a) $S^2 = \mathbf{y}'A\mathbf{y}$ with $A = (I - J_n/n)/(n-1)$, $\mathbf{y} \sim N(\mu\mathbf{1}, \sigma^2 I)$. Since $(I - J_n/n)\mathbf{1} = \mathbf{0}$, $A\mu\mathbf{1} = \mathbf{0}$, so the second term of Theorem 5.4(c) vanishes. Using that $I - J_n/n$ is idempotent with trace $n-1$,

$$\mathrm{Var}\, S^2 = 2\mathrm{tr}\{(\sigma^2 A)^2\} = \frac{2\sigma^4}{(n-1)^2}\mathrm{tr}\Big(I - \frac{J_n}{n}\Big) = \frac{2\sigma^4}{n-1}.$$

(b) $MSE = \mathbf{y}'B\mathbf{y}$ with $B = (I-H)/(n-p-1)$. Since $(I-H)X = 0$, $BX\beta = \mathbf{0}$, and since $I-H$ is idempotent with trace $n-p-1$,

$$\mathrm{Var}(MSE) = 2\sigma^4\mathrm{tr}(B^2) = \frac{2\sigma^4}{(n-p-1)^2}\mathrm{tr}(I-H) = \frac{2\sigma^4}{n-p-1}.$$

(Also from $\mathrm{Var}\{SSE/\sigma^2\} = 2(n-p-1)$.) The relative SE of MSE, $\sqrt{2/(n-p-1)}$, is large for few residual d.f. ■

Chapter 6. Multiple Linear Regression

6.1 The model in matrix form

With p explanatory variables,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \ldots, n,$$

or, stacking the observations,

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

where $\mathbf{y}$ is $n \times 1$, X is the $n \times (p+1)$ **design matrix** whose i-th row is $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$, and $\beta = (\beta_0, \ldots, \beta_p)'$. We assume $n > p+1$ and $\operatorname{rank} X = p+1$. Simple regression is the case $p = 1$, with rows $(1, x_i)$.

6.2 Least squares

Motivation. Matrix notation unifies the formulas: $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$, a projection onto the column space of X.

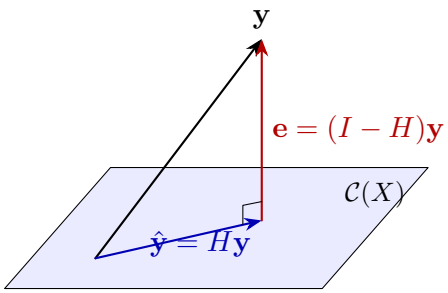


Figure. $\hat{\mathbf{y}}$ *is the orthogonal projection of* $\mathbf{y}$ *onto* $\mathcal{C}(X)$*;* $\mathbf{e} \perp \mathcal{C}(X)$ *is the normal equations* $X^\top \mathbf{e} = \mathbf{0}$.

Theorem 6.1

The least squares estimator minimizing $S(\beta) = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta)$ is the solution of the normal equations $X'X\hat{\beta} = X'\mathbf{y}$:

$$\hat{\beta} = (X'X)^{-1} X'\mathbf{y}.$$

Proof. $S(\beta) = \mathbf{y}'\mathbf{y} - 2\beta' X'\mathbf{y} + \beta' X'X\beta$ has gradient $-2X'\mathbf{y} + 2X'X\beta$ and Hessian $2X'X$, positive definite by full rank ($\mathbf{a}'X'X\mathbf{a} = \|X\mathbf{a}\|^2 > 0$), so the stationary point is the global minimum. Alternatively, for any β,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + \|X(\hat{\beta} - \beta)\|^2,$$

because the cross term $2(\hat{\beta} - \beta)'X'(\mathbf{y} - X\hat{\beta})$ vanishes by the normal equations. ∎

For $p = 1$,

$$X'X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix},$$

which reproduces the formulas of Chapter 3.

Example 6.1

For $n = 6$ observations of (x_1, x_2, y):

x_1	1	2	3	4	5	6
x_2	2	1	4	3	6	5
y	3	6	7	10	12	13

Centring gives $\bar{x}_1 = \bar{x}_2 = 3.5$, $\bar{y} = 8.5$, and the centred cross products $S_{11} = S_{22} = 17.5$, $S_{12} = 14.5$, $S_{1y} = 35.5$, $S_{2y} = 28.5$. The slopes solve

$$\begin{pmatrix} 17.5 & 14.5 \\ 14.5 & 17.5 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} 35.5 \\ 28.5 \end{pmatrix},$$

so, with determinant $17.5^2 - 14.5^2 = 96$,

$$\hat{\beta}_1 = \frac{17.5 \cdot 35.5 - 14.5 \cdot 28.5}{96} = 2.167, \qquad \hat{\beta}_2 = \frac{17.5 \cdot 28.5 - 14.5 \cdot 35.5}{96} = -0.167,$$

and $\hat{\beta}_0 = 8.5 - 3.5(2.167 - 0.167) = 1.5$. Although $S_{2y} > 0$, the coefficient of x_2 is slightly negative given x_1: x_2 adds little beyond x_1 (Section 6.6).

Fitted values $\hat{\mathbf{y}} = X\hat{\beta} = H\mathbf{y}$, **hat matrix** $H = X(X'X)^{-1}X'$, residuals $\mathbf{e} = (I - H)\mathbf{y}$. H is symmetric idempotent, $HX = X$, $\mathrm{tr}H = p + 1$: the projection onto $\mathrm{col}(X)$. So $X'\mathbf{e} = \mathbf{0}$: residuals sum to zero and are orthogonal to every regressor and to $\hat{\mathbf{y}}$.

6.3 Analysis of variance

With $J_n = \mathbf{1}\mathbf{1}'$, $n\bar{y}^2 = \mathbf{y}'(J_n/n)\mathbf{y}$ is the **correction term**. The sums of squares are quadratic forms:

$$SST = \mathbf{y}'\Big(I - \frac{J_n}{n}\Big)\mathbf{y},$$

$$SSE = \mathbf{y}'(I - H)\mathbf{y} = \mathbf{y}'\mathbf{y} - \hat{\beta}'X'\mathbf{y}, \qquad SSR = \mathbf{y}'\Big(H - \frac{J_n}{n}\Big)\mathbf{y} = \hat{\beta}'X'\mathbf{y} - n\bar{y}^2,$$

with $SST = SSR + SSE$. The degrees of freedom are the ranks of the matrices: $n - 1$, p and $n - p - 1$. (SST has $n - 1$ degrees of freedom because the n deviations $y_i - \bar{y}$ satisfy one linear constraint.)

source	SS	df	MS	F_0
regression	SSR	p	$MSR = SSR/p$	MSR/MSE
residual	SSE	$n - p - 1$	$MSE = SSE/(n - p - 1)$	
total	SST	$n - 1$		

Theorem 6.2 (Expected mean squares)

$$E(MSE) = \sigma^2, \qquad E(MSR) = \sigma^2 + \frac{1}{p}\beta'X'\Big(I - \frac{J_n}{n}\Big)X\beta.$$

The second term is ≥ 0, with equality iff $\beta_1 = \cdots = \beta_p = 0$.

Proof. By Theorem 5.4(a) with $E\mathbf{y} = X\beta$, $\mathrm{Var}\,\mathbf{y} = \sigma^2 I$: $B = I - H$ has $\mathrm{tr}B = n - p - 1$, $X'BX = 0$, so $E(SSE) = (n - p - 1)\sigma^2$; $A = H - J_n/n$ has $\mathrm{tr}A = p$ and $X'AX = X'(I - J_n/n)X$ ($HX = X$). And $\beta'X'(I - J_n/n)X\beta = \|(I - J_n/n)X\beta\|^2$ vanishes iff all slopes vanish (centred columns independent). ∎

For $p=1$, $\beta' X'(I-J_n/n)X\beta = \beta_1^2 S_{xx}$, recovering Chapter 5. The **coefficient of determination** is $R^2 = SSR/SST = 1 - SSE/SST$, and $\hat{\sigma}^2 = MSE$ is the **residual mean square**.

6.4 Properties of the estimators

Motivation. $\hat{\boldsymbol{\beta}}$ is linear in $\mathbf{y}$, so its moments are immediate; H shows how fits depend on observations.

Theorem 6.3

(a) $E\hat{\beta} = \beta$ and $\operatorname{Var}\hat{\beta} = \sigma^2(X'X)^{-1}$; writing $(X'X)^{-1} = (c_{ij})$, $\operatorname{Var}\hat{\beta}_j = c_{jj}\sigma^2$ and $\operatorname{Cov}(\hat{\beta}_i, \hat{\beta}_j) = c_{ij}\sigma^2$.

(b) At a point $\mathbf{x} = (1, x_1, \ldots, x_p)'$, $\hat{y} = \mathbf{x}'\hat{\beta}$ estimates $E(y \mid \mathbf{x})$ with $\operatorname{Var}\hat{y} = \mathbf{x}'(X'X)^{-1}\mathbf{x}\,\sigma^2$.

(c) A new observation y_s at $\mathbf{x}$ has $\operatorname{Var}(y_s - \hat{y}) = \{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}\sigma^2$.

Proof. (a) $E\hat{\beta} = (X'X)^{-1}X'X\beta = \beta$ and $\operatorname{Var}\hat{\beta} = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}'\operatorname{Var}(\hat{\beta})\mathbf{x}$.

(c) y_s is independent of $\hat{y}$. ■

For $p=1$: $\operatorname{Var}\hat{\beta}_0 = \sigma^2(1/n + \bar{x}^2/S_{xx})$, $\operatorname{Var}\hat{\beta}_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$, $\operatorname{Var}\hat{y} = \sigma^2\{1/n + (x-\bar{x})^2/S_{xx}\}$. The design enters through $(X'X)^{-1}$: spread x's and avoid near-collinear columns.

6.5 Distribution theory

Motivation. Under normal errors $\hat{\boldsymbol{\beta}}$ and SSE are independent with $SSE/\sigma^2 \sim \chi^2$, giving t and F tests.

Theorem 6.4

Under $\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$:

(a) $SSE/\sigma^2 \sim \chi^2(n-p-1)$;

(b) $SSR/\sigma^2 \sim \chi^2(p, \lambda)$ with $\lambda = \beta' X'(I-J_n/n)X\beta/(2\sigma^2)$;

(c) $SST/\sigma^2 \sim \chi^2(n-1, \lambda)$ with the same λ;

(d) SSR and SSE are independent, and $\hat{\beta}$ is independent of SSE;

(e) $F_0 = MSR/MSE \sim F(p, n-p-1, \lambda)$; under $H_0 : \beta_1 = \cdots = \beta_p = 0$, $F_0 \sim F(p, n-p-1)$.

Proof. All three matrices $I-H$, $H-J_n/n$ and $I-J_n/n$ are symmetric idempotent: for $H-J_n/n$ use $HJ_n = J_nH = J_n$ (since $H\mathbf{1} = \mathbf{1}$, the intercept column) and $J_n^2 = nJ_n$:

$$\Big(H - \frac{J_n}{n}\Big)^2 = H - \frac{J_n}{n} - \frac{J_n}{n} + \frac{J_n}{n} = H - \frac{J_n}{n}.$$

Their ranks (traces) are $n-p-1$, p and $n-1$. Theorem 5.5(b) gives (a)–(c), the noncentralities being $\frac{1}{2\sigma^2}(X\beta)'A(X\beta)$; for $I-H$ it is 0 because $(I-H)X = 0$.

(d) $(H-J_n/n)(I-H) = 0$, so Theorem 5.6(b) gives independence of SSR and SSE. Also $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ and $(X'X)^{-1}X'(I-H) = 0$, so Theorem 5.6(a) applies.

(e) follows from (a), (b), (d). ■

Consequently, with c_{jj} the diagonal of $(X'X)^{-1}$,

$$\frac{\hat\beta_j - \beta_j}{\sqrt{c_{jj} MSE}} \sim t(n-p-1).$$

Example 6.2

In Example 6.1, $SST = \sum (y_i - 8.5)^2 = 73.5$ and $SSR = \hat\beta_1 S_{1y} + \hat\beta_2 S_{2y} = 2.167 \cdot 35.5 - 0.167 \cdot 28.5 = 72.17$, so $SSE = 1.33$ with 3 degrees of freedom and $R^2 = 0.982$. Then $F_0 = (72.17/2)/(1.33/3) = 81.2 > F_{0.05}(2,3) = 9.55$: the regression is highly significant.

6.6 Partial regression: the Frisch–Waugh–Lovell theorem

Partition $X = (X_1, X_2)$, with X_1 $(n \times (q+1))$ containing the intercept and X_2 $(n \times (p-q))$ the remaining columns, and write $\mathbf{y} = X_1\beta_1 + X_2\beta_2 + \epsilon$. Let $H_1 = X_1(X_1'X_1)^{-1}X_1'$.

Theorem 6.5 (Frisch–Waugh–Lovell)

Let $\mathbf{e} = (I - H_1)\mathbf{y}$ be the residuals of regressing $\mathbf{y}$ on X_1, and $X_{2\cdot 1} = (I - H_1)X_2$ the residuals of regressing each column of X_2 on X_1. The least squares coefficient from regressing $\mathbf{e}$ on $X_{2\cdot 1}$,

$$\tilde\beta_2 = (X_{2\cdot 1}'X_{2\cdot 1})^{-1} X_{2\cdot 1}'\mathbf{e},$$

equals the $\hat\beta_2$ part of the full least squares estimator. Moreover the residuals of the two regressions coincide.

Proof [Supplement]. The normal equations of the full model are

$$\begin{pmatrix} X_2'X_2 & X_2'X_1 \\ X_1'X_2 & X_1'X_1 \end{pmatrix} \begin{pmatrix} \hat\beta_2 \\ \hat\beta_1 \end{pmatrix} = \begin{pmatrix} X_2'\mathbf{y} \\ X_1'\mathbf{y} \end{pmatrix}.$$

Multiply on the left by $\begin{pmatrix} I & -X_2'X_1(X_1'X_1)^{-1} \\ 0 & I \end{pmatrix}$. The first block row becomes

$$X_2'(I - H_1)X_2\hat\beta_2 + \{X_2'X_1 - X_2'H_1X_1\}\hat\beta_1 = X_2'(I - H_1)\mathbf{y},$$

and the bracket is 0 ($H_1X_1 = X_1$). So $\hat\beta_2 = \{X_2'(I-H_1)X_2\}^{-1}X_2'(I-H_1)\mathbf{y} = (X_{2\cdot 1}'X_{2\cdot 1})^{-1}X_{2\cdot 1}'\mathbf{e} = \tilde\beta_2$. Residuals: applying $I - H_1$ to $\mathbf{y} - X_1\hat\beta_1 - X_2\hat\beta_2$ (orthogonal to X_1) gives $\mathbf{e} - X_{2\cdot 1}\hat\beta_2$. ■

Interpretation

So $\hat\beta_j$ is the effect of x_j *after adjusting both y and x_j for the other regressors*: the slope of residuals on residuals. It equals the simple slope when x_j is orthogonal to the other centred regressors.

6.7 Selected exercises

Motivation. Independence of $\hat\beta$ and $\mathbf{e}$, R as the maximal correlation, and invariance to parametrization of the column space.

Exercise 6.1 (Joint distribution of $\hat{\beta}$ and e)

Under $\mathbf{y} \sim N_n(X\beta, \sigma^2 I_n)$, find the joint distribution of $(\hat{\beta}', \mathbf{e}')'$, with its mean vector and covariance matrix, and deduce that $\hat{\beta}$ and $\mathbf{e}$ are independent.

Solution.

Step 1. $\begin{pmatrix}\hat{\beta}\\ \mathbf{e}\end{pmatrix} = M\mathbf{y}$ with $M = \begin{pmatrix}(X'X)^{-1}X'\\ I-H\end{pmatrix}$, a fixed $(p+1+n)\times n$ matrix, so the vector is multivariate normal (Section 5.4).

Step 2. (Mean.) $E\hat{\beta} = \beta$ and $E\mathbf{e} = (I-H)X\beta = \mathbf{0}$, since $HX = X$.

Step 3. (Covariance.) $\mathrm{Var}(M\mathbf{y}) = \sigma^2 MM'$. The diagonal blocks are $(X'X)^{-1}X'X(X'X)^{-1} = (X'X)^{-1}$ and $(I-H)(I-H)' = I-H$; the off-diagonal block is $(X'X)^{-1}X'(I-H) = (X'X)^{-1}(X'-X'H) = 0$, because $X'H = (HX)' = X'$. Hence

$$\begin{pmatrix}\hat{\beta}\\ \mathbf{e}\end{pmatrix} \sim N_{p+1+n}\left(\begin{pmatrix}\beta\\ \mathbf{0}\end{pmatrix},\ \sigma^2\begin{pmatrix}(X'X)^{-1} & 0\\ 0 & I-H\end{pmatrix}\right).$$

The distribution is singular, since $I-H$ has rank $n-p-1 < n$.

Step 4. (Independence.) By the mgf characterization (3) of Section 5.4, for $\mathbf{t} = (\mathbf{t}_1', \mathbf{t}_2')'$ the joint mgf is

$$\exp\{\mathbf{t}_1'\beta + \tfrac{\sigma^2}{2}\mathbf{t}_1'(X'X)^{-1}\mathbf{t}_1\}\cdot\exp\{\tfrac{\sigma^2}{2}\mathbf{t}_2'(I-H)\mathbf{t}_2\},$$

so $\hat{\beta} \perp \mathbf{e}$, hence independent of SSE and residual plots (strengthening Theorem 6.4(d)). ■

Exercise 6.2 (Multiple correlation as a maximal correlation)

Let $Y_i = \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$ with centred data ($\sum_i Y_i = 0$, $\sum_i x_{ij} = 0$), X $n\times p$ of full rank, $n > p$; $\hat{Y}_i = \sum_j \hat{\beta}_j x_{ij}$, $SSR = \sum \hat{Y}_i^2$, $SST = \sum Y_i^2$, $\langle \mathbf{a}, \mathbf{b}\rangle = \sum a_i b_i$.
(a) Express SSR and SST through X and $\mathbf{Y}$.
(b) Assuming $SST \neq 0$, prove that

$$\max_{\beta\in\mathbb{R}^p,\ \beta\neq\mathbf{0}} \frac{\langle X\beta, \mathbf{Y}\rangle}{\|X\beta\|\,\|\mathbf{Y}\|} = \sqrt{\frac{SSR}{SST}}.$$

(c) Which β attain the maximum in (b)?

Solution.

(a) $\hat{\mathbf{Y}} = H\mathbf{Y}$ with $H = X(X'X)^{-1}X'$ symmetric idempotent, so

$$SSR = \mathbf{Y}'H'H\mathbf{Y} = \mathbf{Y}'X(X'X)^{-1}X'\mathbf{Y}, \qquad SST = \mathbf{Y}'\mathbf{Y}.$$

(b) For $\beta \neq \mathbf{0}$, $X\beta \neq \mathbf{0}$; as $HX\beta = X\beta$, symmetry of H and Cauchy–Schwarz give

$$\langle X\beta, \mathbf{Y}\rangle = \langle HX\beta, \mathbf{Y}\rangle = \langle X\beta, H\mathbf{Y}\rangle \le \|X\beta\|\,\|H\mathbf{Y}\| = \|X\beta\|\sqrt{SSR}.$$

so the ratio is at most $\sqrt{SSR/SST}$. If $SSR > 0$, $\beta = \hat{\beta}$ attains it; if $SSR = 0$, $H\mathbf{Y} = \mathbf{0}$ and the ratio is always 0.

(c) If $SSR > 0$, equality iff $X\beta = cX\hat{\beta}$, $c > 0$, i.e. $\beta = c\hat{\beta}$; if $SSR = 0$, every $\beta \neq \mathbf{0}$.

For centred data the ratio is the correlation of $\mathbf{Y}$ with $X\beta$: R is the largest correlation of y with a linear combination of regressors, attained by the fitted values. ■

Exercise 6.3 (Reparametrization does not change the fit)

Let A be a nonsingular $(p+1)\times(p+1)$ matrix and $Z = XA$.
(a) Show that $\hat\gamma = A^{-1}\hat\beta$ with $\operatorname{Var}\hat\gamma = \sigma^2 A^{-1}(X'X)^{-1}(A^{-1})'$, and that H, fitted values, residuals, SSE, MSE, R^2 are unchanged.
(b) With $z_j = (x_j - c_j)/d_j$ $(d_j \neq 0)$ and the intercept kept, show that the new slopes are $d_j\hat\beta_j$ and the slope t statistics change at most in sign.

Solution [Supplement].

(a) Z has full column rank, because $Z\mathbf{a} = \mathbf{0}$ gives $XA\mathbf{a} = \mathbf{0}$, so $A\mathbf{a} = \mathbf{0}$ and $\mathbf{a} = \mathbf{0}$. Then

$$\hat\gamma = (A'X'XA)^{-1}A'X'\mathbf{y} = A^{-1}(X'X)^{-1}(A')^{-1}A'X'\mathbf{y} = A^{-1}\hat\beta,$$

and $\operatorname{Var}\hat\gamma = \sigma^2(Z'Z)^{-1} = \sigma^2 A^{-1}(X'X)^{-1}(A^{-1})'$. The hat matrix is

$$Z(Z'Z)^{-1}Z' = XAA^{-1}(X'X)^{-1}(A')^{-1}A'X' = H,$$

so the fitted values $H\mathbf{y}$, the residuals $(I-H)\mathbf{y}$ and SSE coincide; MSE has the same degrees of freedom $n-p-1$, and SST does not involve the design, so R^2 coincides too.

(b) The new design has columns $\mathbf{1}$ and $\mathbf{z}_j = (\mathbf{x}_j - c_j\mathbf{1})/d_j$, so $Z = XA$ with A upper triangular with diagonal $(1, 1/d_1, \ldots, 1/d_p)$, hence nonsingular. By (a) the fitted values agree:

$$\hat\beta_0\mathbf{1} + \sum_j \hat\beta_j\mathbf{x}_j = \hat\gamma_0\mathbf{1} + \sum_j \hat\gamma_j\frac{\mathbf{x}_j - c_j\mathbf{1}}{d_j}.$$

Matching coefficients (independent columns): $\hat\gamma_j = d_j\hat\beta_j$, $\hat\gamma_0 = \hat\beta_0 + \sum_j \hat\beta_j c_j$; likewise $\operatorname{Var}\hat\gamma_j = d_j^2\operatorname{Var}\hat\beta_j$. With the same MSE,

$$\frac{\hat\gamma_j}{\sqrt{d_j^2 c_{jj} MSE}} = \frac{d_j}{|d_j|}\cdot\frac{\hat\beta_j}{\sqrt{c_{jj}MSE}}.$$

So centring and rescaling the regressors (as in Chapter 15) change the units of the slopes but no test or fitted value. ■

Chapter 7. Inference in Multiple Regression I

Throughout, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times (p+1)$ and full column rank. We distinguish two sets of assumptions:

- **(A)** $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n)$, needed for exact t and F inference;
- **(B)** $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I_n$ only, enough for the optimality of least squares.

7.1 The Gauss–Markov theorem

Motivation. Least squares has minimal variance among linear unbiased estimators, without normality.

Theorem 7.1 (Gauss–Markov)

Under (B), among linear unbiased $\tilde{\beta} = A\mathbf{y}$, $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ has the smallest variance: $\operatorname{Var}\tilde{\beta} - \operatorname{Var}\hat{\beta}$ is nonnegative definite; equivalently $\mathbf{c}'\hat{\beta}$ is best for every $\mathbf{c}'\beta$.

Proof [Supplement]. $E(A\mathbf{y}) = AX\beta = \beta$ for all β forces $AX = I$. Write $A = (X'X)^{-1}X' + D$; then $DX = AX - I = 0$. Hence

$$\operatorname{Var}(A\mathbf{y}) = \sigma^2 AA' = \sigma^2\{(X'X)^{-1} + (X'X)^{-1}X'D' + DX(X'X)^{-1} + DD'\} = \sigma^2(X'X)^{-1} + \sigma^2 DD',$$

and DD' is nonnegative definite, with $\mathbf{c}'DD'\mathbf{c} = \|D'\mathbf{c}\|^2$. Equality for all $\mathbf{c}$ holds iff $D = 0$. ■

So $\hat{\beta}$ is the **best linear unbiased estimator** (BLUE), needing only uncorrelated equal-variance errors. If $\operatorname{Var}\epsilon = \sigma^2 V$, $V \neq I$, the BLUE is generalized least squares (Chapter 12).

7.2 Confidence intervals

Motivation. Each coefficient interval describes a predictor's effect holding the others fixed.

Under (A), with $(X'X)^{-1} = (c_{ij})$ and $MSE = SSE/(n-p-1)$ independent of $\hat{\beta}$ (Theorem 6.4):

Theorem 7.2

$100(1-\alpha)\%$ confidence intervals are:

(a) for a coefficient: $\hat{\beta}_j \pm t_{\alpha/2}(n-p-1)\sqrt{c_{jj}MSE}$;

(b) for a linear combination $\mathbf{q}'\beta$: $\mathbf{q}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{q}'(X'X)^{-1}\mathbf{q}\,MSE}$;

(c) for the mean response at $\mathbf{x}$: $\mathbf{x}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{x}'(X'X)^{-1}\mathbf{x}\,MSE}$;

(d) a prediction interval for a new observation at $\mathbf{x}$: $\mathbf{x}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}MSE}$.

With σ^2 known, replace MSE by σ^2 and $t_{\alpha/2}$ by $z_{\alpha/2}$.

Proof. (b) contains (a) and (c) as special cases. $\mathbf{q}'\hat{\beta} \sim N(\mathbf{q}'\beta, \sigma^2\mathbf{q}'(X'X)^{-1}\mathbf{q})$ is independent of $(n-p-1)MSE/\sigma^2 \sim \chi^2(n-p-1)$, so the studentized ratio is $t(n-p-1)$.

(d) adds the variance of the new error. ■

7.3 The general linear hypothesis

Many hypotheses are linear restrictions on β:

$$H_0 : C\beta = \mathbf{m} \quad \text{vs} \quad H_1 : C\beta \neq \mathbf{m},$$

with C a known $k \times (p+1)$ matrix of rank k and $\mathbf{m} \in \mathbb{R}^k$. Examples, for $E(y) = \beta_0 + \beta_1 x_1 + \cdots + \beta_4 x_4$:

- $\beta_1 = \beta_2$ and $\beta_3 = 2\beta_4$: $C = \begin{pmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$, $\mathbf{m} = \mathbf{0}$;
- $\beta_1 = \beta_2 = \beta_3 = \beta_4$: three rows $(0, 1, -1, 0, 0)$, $(0, 0, 1, -1, 0)$, $(0, 0, 0, 1, -1)$;
- $\beta_j = 0$: $C = \mathbf{e}_j'$, the j-th unit row.

There are two equivalent routes: minimize SSE under the restriction, or reparametrize to a reduced model.

7.3.1 Method I: constrained least squares

Theorem 7.3 (Restricted least squares)

The minimizer of $\|\mathbf{y} - X\beta\|^2$ subject to $C\beta = \mathbf{m}$ is

$$\tilde{\beta} = \hat{\beta} - (X'X)^{-1}C'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}),$$

and its residual sum of squares is

$$SSE(\tilde{\beta}) = SSE + Q, \qquad Q = (C\hat{\beta} - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}).$$

Proof [Supplement]. With a Lagrange multiplier vector $\lambda \in \mathbb{R}^k$, minimize $L = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta) + 2\lambda'(C\beta - \mathbf{m})$. Setting derivatives to zero:

$$X'X\tilde{\beta} + C'\lambda = X'\mathbf{y}, \qquad C\tilde{\beta} = \mathbf{m}.$$

The first gives $\tilde{\beta} = \hat{\beta} - (X'X)^{-1}C'\lambda$; the second then gives $\lambda = \{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m})$ (invertible, C of full row rank). The decomposition below shows this is the constrained minimum: for any β,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + (\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta),$$

since $X'(\mathbf{y} - X\hat{\beta}) = \mathbf{0}$. With $\beta = \tilde{\beta}$ and $\hat{\beta} - \tilde{\beta} = (X'X)^{-1}C'\lambda$,

$$(\hat{\beta} - \tilde{\beta})'X'X(\hat{\beta} - \tilde{\beta}) = \lambda'C(X'X)^{-1}C'\lambda = (C\hat{\beta} - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}) = Q.$$

For any feasible β, $(\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta) \geq Q$ (it is the squared distance, in the $X'X$ metric, from $\hat{\beta}$ to the affine set $\{C\beta = \mathbf{m}\}$, attained at $\tilde{\beta}$), so $\tilde{\beta}$ is optimal. ■

Theorem 7.4 (F test of the general linear hypothesis)

Under (A),

$$\frac{Q}{\sigma^2} \sim \chi^2(k, \lambda), \qquad \lambda = \frac{(C\beta - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\beta - \mathbf{m})}{2\sigma^2},$$

Q is independent of SSE, and

$$F_0 = \frac{Q/k}{SSE/(n-p-1)} = \frac{Q}{k \cdot MSE} \sim F(k, n-p-1, \lambda).$$

Under H_0, $\lambda = 0$, and the test rejects when $F_0 > F_\alpha(k, n-p-1)$.

Proof. $C\hat{\beta} - \mathbf{m} \sim N(C\beta - \mathbf{m}, \sigma^2 C(X'X)^{-1}C')$ and Q is its quadratic form with $W = \{C(X'X)^{-1}C'\}^{-1}$; $W \cdot C(X'X)^{-1}C' = I_k$ is idempotent, so Theorem 5.5 applies. Q depends on $\hat{\beta}$ only, independent of SSE (Theorem 6.4). ■

7.3.2 Method II: full and reduced models

Substitute the restriction into the model to get a **reduced model** with $p+1-k$ free parameters; fit it and compute

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/k}{SSE(F)/(n-p-1)},$$

with $df_F = n-p-1$, $df_R = n-p-1+k$ (a nonzero $\mathbf{m}$ becomes an offset). Since $SSE(R) = SSE(\tilde{\beta})$, Theorem 7.3 gives $SSE(R) - SSE(F) = Q$: *both methods give the same statistic.*

For a model without intercept, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times p$, the same holds with $n-p$ in place of $n-p-1$.

Example 7.1

A no-intercept model $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$ is fitted to

x_1	1	2	0	1	3
x_2	0	1	1	2	1
y	3	7	2	5	10

Test $H_0 : \beta_1 = 2\beta_2$. Here

$$X'X = \begin{pmatrix} 15 & 7 \\ 7 & 7 \end{pmatrix}, \qquad X'\mathbf{y} = \begin{pmatrix} 52 \\ 29 \end{pmatrix}, \qquad \hat{\beta} = \frac{1}{56}\begin{pmatrix} 7 & -7 \\ -7 & 15 \end{pmatrix}\begin{pmatrix} 52 \\ 29 \end{pmatrix} = \begin{pmatrix} 2.875 \\ 1.268 \end{pmatrix},$$

$\mathbf{y}'\mathbf{y} = 187$, and $SSE = 187 - \hat{\beta}'X'\mathbf{y} = 0.732$ with 3 degrees of freedom, so $MSE = 0.244$. *Method I*: $C = (1, -2)$, $C\hat{\beta} = 0.339$, $C(X'X)^{-1}C' = \frac{7+28+60}{56} = 1.696$, so $Q = 0.339^2/1.696 = 0.068$ and $F_0 = 0.068/0.244 = 0.28$. *Method II*: under H_0, $y = \beta_2(2x_1 + x_2) + \epsilon$; with $z = 2x_1 + x_2 = (2, 5, 1, 4, 7)$, $\sum z^2 = 95$ and $\sum zy = 133$, so $SSE(R) = 187 - 133^2/95 = 0.800$ and $\{SSE(R) - SSE\}/MSE = 0.068/0.244 = 0.28$. Both give $F_0 = 0.28 < F_{0.05}(1, 3) = 10.13$: H_0 is not rejected.

7.4 Properties of residuals

Motivation. Residuals are correlated with unequal variances $\sigma^2(1 - h_{ii})$, motivating Chapter 10.

Theorem 7.5

For the fitted model with intercept:

(a) $\sum_i e_i = 0$, $\sum_i x_{ij} e_i = 0$ for each j, and $\sum_i \hat{y}_i e_i = 0$;

(b) under (A), $\mathbf{e} = (I - H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I - H))$; in particular $\operatorname{Var} e_i = \sigma^2(1 - h_{ii})$ and $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$.

Proof. (a) is $X'\mathbf{e} = \mathbf{0}$ together with $\hat{\mathbf{y}}'\mathbf{e} = \hat{\beta}'X'\mathbf{e}$.

(b) $E\mathbf{e} = (I - H)X\beta = \mathbf{0}$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I - H)(I - H)' = \sigma^2(I - H)$. ■

So residuals have correlation $-h_{ij}/\sqrt{(1 - h_{ii})(1 - h_{jj})}$ and design-dependent variances. Basic checks plot them against $\hat{y}_i$, each x_{ij} and time order:

- a horizontal band: no problem;
- a funnel: nonconstant variance; use weighted least squares or transform y;
- a linear trend against x_j: a computational error, since (a) forces zero correlation;
- a curve: add a quadratic term (or other function) of x_j, or transform.

7.5 Selected exercises

Motivation. The general linear hypothesis for a constrained mean and for weights summing to one, and the bias–variance effect of a restriction.

Exercise 7.1 (An F statistic under a side condition)

Let $\mathbf{y} = \theta + \epsilon$ with $\mathbf{y}, \theta, \epsilon \in \mathbb{R}^4$ and $\epsilon \sim N_4(\mathbf{0}, \sigma^2 I_4)$, where θ is known to satisfy $\theta_1 + \theta_2 + \theta_3 + \theta_4 = 0$. Show that the F statistic for $H_0 : \theta_1 = \theta_3$ is

$$F_0 = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2},$$

and state its null distribution.

Solution.

Step 1. (The model as a regression.) $\theta \in \Omega = \{\theta : \mathbf{1}'\theta = 0\}$, a 3-dimensional subspace; with the basis $(1, -1, 0, 0)'$, $(0, 1, -1, 0)'$, $(0, 0, 1, -1)'$ as columns of X, $\mathbf{y} = X\beta + \epsilon$ without intercept, and H_0 is one restriction. Method II (Section 7.3): $F_0 = \{SSE(R) - SSE(F)\}/\{SSE(F)/(4 - 3)\}$; for a mean in subspace L, $SSE = \|\mathbf{y} - P_L\mathbf{y}\|^2$ (Theorem 6.1).

Step 2. (Full model.) Ω is the orthogonal complement of $\mathbf{1}$, so $P_\Omega\mathbf{y} = \mathbf{y} - \bar{y}\mathbf{1}$ and

$$SSE(F) = \|\bar{y}\mathbf{1}\|^2 = 4\bar{y}^2 = \frac{(y_1 + y_2 + y_3 + y_4)^2}{4}.$$

Step 3. (Reduced model.) Let $\mathbf{c} = (1, 0, -1, 0)'$. Then $\mathbf{c} \in \Omega$, and the reduced mean space is $\omega = \{\theta \in \Omega : \mathbf{c}'\theta = 0\}$, the orthogonal complement of $\mathbf{c}$ inside Ω. Hence $\Omega = \omega \oplus \operatorname{span}(\mathbf{c})$ orthogonally and $P_\Omega = P_\omega + P_\mathbf{c}$, with $P_\mathbf{c}\mathbf{y} = (\mathbf{c}'\mathbf{y}/\|\mathbf{c}\|^2)\mathbf{c}$. Since $\mathbf{y} - P_\Omega\mathbf{y} \perp \Omega \ni P_\mathbf{c}\mathbf{y}$, Pythagoras gives

$$SSE(R) = \|\mathbf{y} - P_\Omega\mathbf{y} + P_\mathbf{c}\mathbf{y}\|^2 = SSE(F) + \frac{(\mathbf{c}'\mathbf{y})^2}{\|\mathbf{c}\|^2} = SSE(F) + \frac{(y_1 - y_3)^2}{2}.$$

Step 4. Therefore

$$F_0 = \frac{(y_1 - y_3)^2/2}{(y_1 + y_2 + y_3 + y_4)^2/4} = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2},$$

and under H_0, $F_0 \sim F(1, 1)$ by Theorem 7.4 (with $n - p$ in place of $n - p - 1$ for a model without intercept, here $4 - 3 = 1$). ■

Exercise 7.2 (Weights that sum to one)

In a survey, n respondents report scores $X_{i1}, \dots, X_{ip}$ on p items and an overall score Y_i. The model is

$$Y_i = \sum_{j=1}^{p} \omega_j X_{ij} + \epsilon_i, \qquad \sum_{j=1}^{p} \omega_j = 1, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

with X $n \times p$ of full rank, $n > p$ (negative weights allowed). Derive the LSE of ω and an unbiased estimator of σ^2, with their distributions.

Solution. Write $G = (X'X)^{-1}$, $\hat{\omega} = GX'\mathbf{Y}$ (the unconstrained estimator), $\mathbf{g} = G\mathbf{1}$ and $s = \mathbf{1}'G\mathbf{1} > 0$.

Step 1. (Estimator.) Theorem 7.3 with $C = \mathbf{1}'$ ($k = 1$) and $\mathbf{m} = 1$ gives

$$\tilde{\omega} = \hat{\omega} - \frac{\mathbf{1}'\hat{\omega} - 1}{s}\mathbf{g} = \Big(I - \frac{\mathbf{g}\mathbf{1}'}{s}\Big)\hat{\omega} + \frac{\mathbf{g}}{s},$$

and indeed $\mathbf{1}'\tilde{\omega} = \mathbf{1}'\hat{\omega} - (\mathbf{1}'\hat{\omega} - 1) = 1$.

Step 2. (Reduced model.) Substituting $\omega_p = 1 - \sum_{j<p} \omega_j$,

$$Y_i - X_{ip} = \sum_{j<p} \omega_j (X_{ij} - X_{ip}) + \epsilon_i,$$

an unconstrained regression of $\mathbf{Y} - \mathbf{X}_p$ on $Z = XB$, B the $p \times (p-1)$ matrix with columns $\mathbf{e}_j - \mathbf{e}_p$; Z has full column rank $p-1$. As $(\omega_1, \dots, \omega_{p-1}) \mapsto \omega$ is a bijection onto the constraint set preserving $\|\mathbf{Y} - X\omega\|^2$, $\tilde{\omega}$ is this regression's estimator ($\tilde{\omega}_p = 1 - \sum_{j<p} \tilde{\omega}_j$), with residual sum of squares $SSE(\tilde{\omega})$ on $n - p + 1$ d.f.

Step 3. (σ^2.) Theorem 6.4 for the reduced regression gives $SSE(\tilde{\omega})/\sigma^2 \sim \chi^2(n-p+1)$, independent of $\tilde{\omega}$. Thus

$$\tilde{\sigma}^2 = \frac{\|\mathbf{Y} - X\tilde{\omega}\|^2}{n - p + 1}$$

is unbiased, and $(n - p + 1)\tilde{\sigma}^2/\sigma^2 \sim \chi^2(n - p + 1)$.

Step 4. (Distribution of $\tilde{\omega}$.) $\tilde{\omega}$ is an affine function of $\hat{\omega} \sim N_p(\omega, \sigma^2 G)$, hence normal. Since $\mathbf{1}'\omega = 1$, $E\tilde{\omega} = \omega - \frac{1-1}{s}\mathbf{g} = \omega$. With $M = I - \mathbf{g}\mathbf{1}'/s$,

$$\operatorname{Var}\tilde{\omega} = \sigma^2 MGM' = \sigma^2\Big(G - \frac{\mathbf{g}\mathbf{g}'}{s} - \frac{\mathbf{g}\mathbf{g}'}{s} + \frac{\mathbf{g}(\mathbf{1}'G\mathbf{1})\mathbf{g}'}{s^2}\Big) = \sigma^2\Big(G - \frac{\mathbf{g}\mathbf{g}'}{s}\Big).$$

So $\tilde{\omega} \sim N_p(\omega, \sigma^2(G - \mathbf{g}\mathbf{g}'/s))$, a singular normal distribution concentrated on the hyperplane $\mathbf{1}'\omega = 1$: indeed $(G - \mathbf{g}\mathbf{g}'/s)\mathbf{1} = \mathbf{g} - \mathbf{g} = \mathbf{0}$. Replacing σ^2 by $\tilde{\sigma}^2$ gives t intervals for each ω_j with $n - p + 1$ degrees of freedom. ■

Exercise 7.3 (Bias and variance of restricted least squares)

Under assumption (B), let $\tilde{\beta}$ be the restricted estimator of Theorem 7.3, and $G = (X'X)^{-1}$, $W = \{CGC'\}^{-1}$.
(a) Show that $E\tilde{\beta} = \beta - GC'W(C\beta - \mathbf{m})$; in particular $\tilde{\beta}$ is unbiased when the restriction is true.
(b) Show that $\operatorname{Var}\tilde{\beta} = \sigma^2(G - GC'WCG)$, so that $\operatorname{Var}\hat{\beta} - \operatorname{Var}\tilde{\beta}$ is nonnegative definite whether or not the restriction is true.

Solution [Supplement].

(a) By Theorem 7.3, $\tilde{\beta} = M\hat{\beta} + GC'W\mathbf{m}$ with $M = I - GC'WC$. Since $E\hat{\beta} = \beta$,

$$E\tilde{\beta} = \beta - GC'WC\beta + GC'W\mathbf{m} = \beta - GC'W(C\beta - \mathbf{m}),$$

which equals β when $C\beta = \mathbf{m}$.

(b) $\operatorname{Var}\tilde{\beta} = M(\sigma^2 G)M'$. Using $G' = G$, $W' = W$ and $W(CGC')W = W$,

$$\begin{aligned} MGM' &= (G - GC'WCG)(I - C'WCG) \\ &= G - GC'WCG - GC'WCG + GC'W(CGC')WCG = G - GC'WCG. \end{aligned}$$

So $\operatorname{Var}\hat{\beta} - \operatorname{Var}\tilde{\beta} = \sigma^2(CG)'W(CG)$ is nonnegative definite (W positive definite): a correct restriction reduces variance without bias; a false one adds the bias of (a). ■

Chapter 8. Inference in Multiple Regression II

8.1 Extra sums of squares

Adding regressors increases SSR; the increase, the **extra sum of squares**, measures their contribution *given* those present.

For the full model $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$, write

$$SS(\hat\beta_0, \ldots, \hat\beta_p) = \hat\beta' X'\mathbf{y} \quad (df = p+1), \qquad SS(\hat\beta_0) = n\bar y^2 \quad (df = 1),$$

$$SS(\hat\beta_1, \ldots, \hat\beta_p \mid \hat\beta_0) = SS(\hat\beta_0, \ldots, \hat\beta_p) - SS(\hat\beta_0) = SSR \quad (df = p).$$

If a submodel uses only $x_1, \ldots, x_q$ ($q < p$) with fit $\tilde\beta$, the extra sum of squares of $x_{q+1}, \ldots, x_p$ given $x_1, \ldots, x_q$ is

$$SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q) = SS(\hat\beta) - SS(\tilde\beta) = SSR(\text{full}) - SSR(\text{sub}) = SSE(\text{sub}) - SSE(\text{full}),$$

with $p - q$ degrees of freedom. In matrix form, partition $\beta = (\beta_1', \beta_2')'$ with $\beta_1 = (\beta_0, \ldots, \beta_q)'$, $X = (X_1, X_2)$, and let $H_1 = X_1(X_1'X_1)^{-1}X_1'$, $H = X(X'X)^{-1}X'$:

$$SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H - H_1)\mathbf{y}.$$

Theorem 8.1

(a) $H - H_1$ is symmetric idempotent of rank $p - q$, and $(H - H_1)(I - H) = 0$.

(b) Under normal errors, $SS(\beta_2 \mid \beta_1)/\sigma^2 \sim \chi^2(p-q, \lambda)$ with $\lambda = \beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$, independent of SSE.

(c) Hence, under $H_0 : \beta_{q+1} = \cdots = \beta_p = 0$,

$$F_0 = \frac{SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q)/(p-q)}{MSE} \sim F(p-q, n-p-1).$$

Proof. (a) $HX = X$ gives $HX_1 = X_1$, hence $HH_1 = H_1$ and (transposing) $H_1H = H_1$. Then $(H-H_1)^2 = H - H_1 - H_1 + H_1 = H - H_1$, and $\operatorname{tr}(H - H_1) = (p+1) - (q+1)$. Also $(H - H_1)(I - H) = H - H - H_1 + H_1 = 0$.

(b) By Theorem 5.5(b) the noncentrality is $\frac{1}{2\sigma^2}(X\beta)'(H - H_1)X\beta$. Since $(H - H_1)X_1 = 0$ and $(H - H_1)X_2 = (I - H_1)X_2$, this equals $\beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$. Independence follows from (a) and Theorem 5.6.

(c) Under H_0, $\lambda = 0$. ∎

This is the same test as the general linear hypothesis with $C = (0, I_{p-q})$ (Chapter 7), since $SSE(\text{sub}) - SSE(\text{full}) = Q$.

Including the intercept in SSR gives the uncorrected table: $SSR = \hat\beta' X'\mathbf{y}$ ($p+1$ d.f.), $SST = \mathbf{y}'\mathbf{y}$ (n), and $MSR/MSE \sim F(p+1, n-p-1)$ tests $\beta = \mathbf{0}$.

8.2 Partial and sequential F tests

Partial F test. The contribution of a single regressor x_j given all others is $SS(\hat\beta_j \mid \text{all others})$, with one degree of freedom. The test of $H_0 : \beta_j = 0$,

$$F_0 = \frac{SS(\hat\beta_j \mid \hat\beta_0, \ldots, \hat\beta_{j-1}, \hat\beta_{j+1}, \ldots, \hat\beta_p)}{MSE} \sim F(1, n-p-1),$$

is the **partial F test**. It is equivalent to the t test, since

$$SS(\hat\beta_j \mid \text{others}) = \frac{\hat\beta_j^2}{c_{jj}}, \qquad F_0 = \Big(\frac{\hat\beta_j}{\sqrt{c_{jj}MSE}}\Big)^2 = t_0^2.$$

(Theorem 7.3 with $C = \mathbf{e}_j'$: $Q = \hat\beta_j^2/c_{jj}$.) A non-significant partial F means x_j adds little *given the others*, not that it is unrelated to y; with correlated regressors all partial Fs can be small while the overall F is large.

Sequential F tests. Enter regressors one at a time, testing each given those already in:

$$SS(\hat\beta_1 \mid \hat\beta_0), \quad SS(\hat\beta_2 \mid \hat\beta_0, \hat\beta_1), \quad \ldots, \quad SS(\hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_{p-1}).$$

These sum to SSR; each is compared with the stage (or full-model) MSE. The decomposition depends on order unless the centred regressors are orthogonal, and needs an unbiased MSE (omitted variables without effect).

8.3 Joint confidence regions and simultaneous intervals

Motivation. Joint regions, Bonferroni and Scheffé intervals control the error rate of all statements at once.

Theorem 8.2 (Joint confidence region)

Under normal errors,

$$\frac{(\hat\beta-\beta)'X'X(\hat\beta-\beta)}{(p+1)MSE} \sim F(p+1, n-p-1),$$

so $\{\beta : (\hat\beta-\beta)'X'X(\hat\beta-\beta) \le (p+1)MSE\, F_\alpha(p+1,n-p-1)\}$ is a $100(1-\alpha)\%$ confidence ellipsoid for β.

Proof. $\hat\beta - \beta = (X'X)^{-1}X'\epsilon$, so $(\hat\beta-\beta)'X'X(\hat\beta-\beta) = \epsilon' H \epsilon$, and $\epsilon' H\epsilon/\sigma^2 \sim \chi^2(p+1)$. It is independent of $SSE = \epsilon'(I-H)\epsilon$, since $H(I-H)=0$. ∎

Separate 95% intervals for $\beta_0, \ldots, \beta_p$ do not have joint coverage 95%. Two remedies:

Theorem 8.3 (Bonferroni and Scheffé intervals)

(a) (Bonferroni) The intervals $\hat\beta_j \pm t_{\alpha/(2(p+1))}(n-p-1)\sqrt{c_{jj}MSE}$, $j = 0, \ldots, p$, cover all β_j simultaneously with probability at least $1-\alpha$.

(b) (Scheffé) With probability $1-\alpha$, simultaneously for all $\mathbf{a} \in \mathbb{R}^{p+1}$,

$$\mathbf{a}'\beta \in \mathbf{a}'\hat\beta \pm \sqrt{(p+1)F_\alpha(p+1,n-p-1)}\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}.$$

Proof. (a) If A_j is the event that the j-th interval covers, $P(A_j^c) = \alpha/(p+1)$, and $P(\bigcap A_j) \ge 1 - \sum P(A_j^c) = 1-\alpha$.

(b) By the Cauchy–Schwarz inequality in the inner product defined by $(X'X)^{-1}$,

$$\max_{\mathbf{a}\neq\mathbf{0}} \frac{\{\mathbf{a}'(\hat\beta-\beta)\}^2}{\mathbf{a}'(X'X)^{-1}\mathbf{a}} = (\hat\beta-\beta)'X'X(\hat\beta-\beta),$$

so the event in (b) is exactly the joint confidence region of Theorem 8.2. ■

$\mathbf{a} = \mathbf{x} = (1, x_1, \ldots, x_p)'$ in (b) gives a **confidence band** for the whole surface. Bonferroni is shorter for few coefficients; Scheffé covers infinitely many combinations.

8.4 Testing equality of regression models

Two populations follow

$$\mathbf{y}_1 = X_1\beta_1 + \epsilon_1, \qquad \mathbf{y}_2 = X_2\beta_2 + \epsilon_2,$$

with n_1 and n_2 observations, design matrices of size $n_i \times (p+1)$, and independent $N(\mathbf{0}, \sigma^2 I)$ errors with a *common* σ^2. To test $H_0 : \beta_1 = \beta_2$:

(1) Full model: fit each separately, $SSE(F) = SSE_1 + SSE_2$, with $n-2(p+1)$ degrees of freedom ($n = n_1 + n_2$).

(2) Reduced model: stack $\mathbf{y} = (\mathbf{y}_1', \mathbf{y}_2')'$ and $X = (X_1', X_2')'$ and fit one β_0: $SSE(R)$, with $n-(p+1)$ degrees of freedom.

(3) Under H_0,

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/(p+1)}{SSE(F)/(n-2p-2)} \sim F(p+1, n-2p-2).$$

The reduced model is the full model (design $\mathrm{diag}(X_1, X_2)$, parameter $(\beta_1', \beta_2')'$) under $p+1$ restrictions $\beta_1 - \beta_2 = \mathbf{0}$: a general linear hypothesis. For k groups the d.f. are $(k-1)(p+1)$ and $n - k(p+1)$.

The test assumes equal error variances; with $k = 2$ this can be checked first by $F = MSE_1/MSE_2 \sim F(n_1 - p - 1, n_2 - p - 1)$.

8.5 Selected exercises

Motivation. The equality test via coefficient differences, partial F through R^2, order-free sequential sums, and Scheffé vs t intervals.

Exercise 8.1 (Equality of two regressions via $\hat\beta_1 - \hat\beta_2$)

In the setting of Section 8.4, let $\hat\beta_1, \hat\beta_2$ be the separate least squares estimators, $G_i = (X_i'X_i)^{-1}$, $\hat\Delta = \hat\beta_1 - \hat\beta_2$ and $MSE = (SSE_1 + SSE_2)/(n-2p-2)$.
(a) Find the distribution of $\hat\Delta$, and show that $(n-2p-2)MSE/\sigma^2 \sim \chi^2(n-2p-2)$, independent of $\hat\Delta$.
(b) Show that under $H_0 : \beta_1 = \beta_2$,

$$F_1 = \frac{\hat\Delta'(G_1 + G_2)^{-1}\hat\Delta/(p+1)}{MSE} \sim F(p+1, n-2p-2).$$

(c) Show that F_1 equals the statistic F_0 of Section 8.4.

Solution.

(a) $\hat\beta_i \sim N_{p+1}(\beta_i, \sigma^2 G_i)$ (Theorem 6.3), independent across samples, so $\hat\Delta \sim N_{p+1}(\beta_1 - \beta_2, \sigma^2(G_1 + G_2))$; by Theorem 6.4, $\hat\beta_1, SSE_1, \hat\beta_2, SSE_2$ are mutually independent, so $(SSE_1 + SSE_2)/\sigma^2 \sim \chi^2(n-2p-2)$ independent of $\hat\Delta$.

(b) $\Sigma = \sigma^2(G_1 + G_2)$ is positive definite. Under H_0, $\hat{\Delta} \sim N(\mathbf{0}, \Sigma)$, and with $A = \Sigma^{-1}$ we have $A\Sigma = I$, which is idempotent of rank $p+1$; Theorem 5.5(a) gives $\hat{\Delta}'\Sigma^{-1}\hat{\Delta} \sim \chi^2(p+1)$. Dividing by $(n-2p-2)MSE/\sigma^2$ and using (a),

$$F_1 = \frac{\hat{\Delta}'\Sigma^{-1}\hat{\Delta}/(p+1)}{\{(n-2p-2)MSE/\sigma^2\}/(n-2p-2)} \sim F(p+1, n-2p-2).$$

(c) Write the full model as one regression with design $X_F = \operatorname{diag}(X_1, X_2)$ and parameter $\mathbf{b} = (\beta_1', \beta_2')'$. Then $X_F'X_F = \operatorname{diag}(X_1'X_1, X_2'X_2)$, so the normal equations decouple, $\hat{\mathbf{b}} = (\hat{\beta}_1', \hat{\beta}_2')'$ and $SSE(F) = SSE_1 + SSE_2$. The hypothesis is $C\mathbf{b} = \mathbf{0}$ with $C = (I_{p+1}, -I_{p+1})$, and

$$C(X_F'X_F)^{-1}C' = G_1 + G_2, \qquad C\hat{\mathbf{b}} = \hat{\Delta}, \qquad Q = \hat{\Delta}'(G_1+G_2)^{-1}\hat{\Delta}.$$

By Theorem 7.3 (Method II), $SSE(R) - SSE(F) = Q$, so $F_0 = \{Q/(p+1)\}/MSE = F_1$. ■

Exercise 8.2 (Extra sums of squares through R^2)

Let the full model (with intercept) have p regressors and coefficient of determination R_F^2, and the submodel the first q of them, with R_R^2.
(a) Show that the F statistic of Theorem 8.1(c) equals

$$F_0 = \frac{(R_F^2 - R_R^2)/(p-q)}{(1-R_F^2)/(n-p-1)}.$$

(b) Show that $R_F^2 \geq R_R^2$, with equality if and only if $\hat{\beta}_{q+1} = \cdots = \hat{\beta}_p = 0$ in the full fit.

Solution [Supplement].

(a) Both models contain the intercept and are fitted to the same $\mathbf{y}$, so they share SST, and $SSE = SST(1-R^2)$ for each. Hence $SS(\beta_2 \mid \beta_1) = SSE(\text{sub}) - SSE(\text{full}) = SST(R_F^2 - R_R^2)$ and $MSE = SST(1-R_F^2)/(n-p-1)$; SST cancels in the ratio.

(b) $R_F^2 - R_R^2 = \mathbf{y}'(H - H_1)\mathbf{y}/SST = \|(H-H_1)\mathbf{y}\|^2/SST \geq 0$, because $H - H_1$ is symmetric idempotent (Theorem 8.1(a)). By Theorem 6.5, the full-model residual is $(I-H)\mathbf{y} = (I-H_1)\mathbf{y} - X_{2\cdot1}\hat{\beta}_2$, so

$$(H-H_1)\mathbf{y} = (I-H_1)\mathbf{y} - (I-H)\mathbf{y} = X_{2\cdot1}\hat{\beta}_2.$$

$X_{2\cdot1} = (I-H_1)X_2$ has full column rank: if $X_{2\cdot1}\mathbf{a} = \mathbf{0}$, then $X_2\mathbf{a} = H_1X_2\mathbf{a}$ lies in the column space of X_1, contradicting the full rank of X unless $\mathbf{a} = \mathbf{0}$. Therefore $R_F^2 = R_R^2$ iff $X_{2\cdot1}\hat{\beta}_2 = \mathbf{0}$ iff $\hat{\beta}_2 = \mathbf{0}$. ■

Exercise 8.3 (When the order of entry does not matter)

Partition $X = (X_1, X_2)$ as in Section 8.1, X_1 containing the intercept, and suppose $X_1'X_2 = 0$ (for instance, the columns of X_2 are centred and uncorrelated with those of X_1). Let $H_2 = X_2(X_2'X_2)^{-1}X_2'$.
(a) Show that $H = H_1 + H_2$, and hence $SS(\beta_2 \mid \beta_1) = \mathbf{y}'H_2\mathbf{y}$, which does not involve X_1.
(b) Show that $\hat{\beta}_2 = (X_2'X_2)^{-1}X_2'\mathbf{y}$.
(c) Deduce that sequential sums of squares of mutually orthogonal centred blocks do not depend on the order.

Solution [Supplement].

(a) $H_1H_2 = X_1(X_1'X_1)^{-1}(X_1'X_2)(X_2'X_2)^{-1}X_2' = 0 = H_2H_1$, so $P = H_1 + H_2$ is an orthogonal projection whose range contains X_1, X_2 and lies in $\operatorname{col}(X)$: $P = H$, and $SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H-H_1)\mathbf{y} = \mathbf{y}'H_2\mathbf{y}$.

(b) $(I - H_1)X_2 = X_2 - X_1(X_1'X_1)^{-1}X_1'X_2 = X_2$, so $X_{2\cdot 1} = X_2$, and by Theorem 6.5, $\hat{\beta}_2 = (X_2'X_2)^{-1}X_2'(I - H_1)\mathbf{y} = (X_2'X_2)^{-1}X_2'\mathbf{y}$, using $X_2'H_1 = 0$.

(c) For centred blocks with $Z_j'Z_k = 0$ and $\mathbf{1}'Z_j = \mathbf{0}$, earlier blocks plus intercept form an X_1 orthogonal to Z_j, so by (a) its sequential sum is $\mathbf{y}'H_{Z_j}\mathbf{y}$ in every order; these sum to SSR. ■

Exercise 8.4 (Scheffé intervals are wider than t intervals)

Let $\nu = n - p - 1$. Show that $(p+1)F_\alpha(p+1, \nu) \geq t_{\alpha/2}(\nu)^2$, so that for a single combination $\mathbf{a}'\beta$ the Scheffé interval of Theorem 8.3(b) contains the t interval of Theorem 7.2(b).

Solution [Supplement].

Step 1. Fix $\mathbf{a} \neq \mathbf{0}$ and let $T_{\mathbf{a}} = \mathbf{a}'(\hat{\beta} - \beta)/\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}$. By Theorem 7.2, $T_{\mathbf{a}} \sim t(\nu)$, so $T_{\mathbf{a}}^2 \sim F(1, \nu)$, whose distribution function is continuous and strictly increasing on $(0, \infty)$.

Step 2. By the proof of Theorem 8.3(b), $\sup_{\mathbf{b} \neq \mathbf{0}} T_{\mathbf{b}}^2 = (\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta)/MSE$, and by Theorem 8.2 this is at most $c = (p+1)F_\alpha(p+1, \nu)$ with probability $1 - \alpha$.

Step 3. Since $T_{\mathbf{a}}^2 \leq \sup_{\mathbf{b}} T_{\mathbf{b}}^2$, $P(T_{\mathbf{a}}^2 \leq c) \geq 1 - \alpha = P(T_{\mathbf{a}}^2 \leq t_{\alpha/2}(\nu)^2)$. By the strict monotonicity in Step 1, $c \geq t_{\alpha/2}(\nu)^2$. The half-widths are $\sqrt{c}$ and $t_{\alpha/2}(\nu)$ times the same standard error, which proves the claim. ■

Chapter 10. Regression Diagnostics

10.1 What can go wrong

Fitting $\mathbf{y} = X\beta + \epsilon$, $\epsilon \sim N(\mathbf{0}, \sigma^2 I)$, by $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ is justified only if the model is adequate. **Regression diagnostics** look for:

(1) violations of the error assumptions (normality, independence, equal variance), studied through residuals;

(2) a wrong functional form, remedied by transforming y or the x_j (Chapters 5 and 11);

(3) near-singularity of $X'X$ (multicollinearity, Chapter 15);

(4) **high-leverage points**: observations whose $\mathbf{x}_i$ is far from the bulk of the design;

(5) **outliers**: observations whose y_i is far from its fitted value;

(6) **influential observations**: observations whose deletion changes the fit substantially.

Types (4)–(6) need not coincide: a high-leverage point can lie on the line, and a central outlier may barely move the fit.

10.2 The hat matrix and leverage

Motivation. The hat-matrix diagonal measures how far a point lies in x-space, flagging leverage before looking at y.

The fitted values are $\hat{\mathbf{y}} = H\mathbf{y}$ with $H = X(X'X)^{-1}X' = (h_{ij})$, where $h_{ij} = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_j$ and $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$ is the i-th row of X.

Theorem 10.1 (Properties of the hat matrix)

(a) H is symmetric and idempotent with $\operatorname{rank} H = \operatorname{tr} H = \sum_i h_{ii} = p + 1$.

(b) $h_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$; hence $0 \le h_{ii} \le 1$ and $|h_{ij}| \le \frac{1}{2}$ for $i \neq j$.

(c) $HX = X$. If the model has an intercept, $H\mathbf{1} = \mathbf{1}$, i.e. every row of H sums to 1, and $h_{ii} \ge 1/n$.

(d) With an intercept, $h_{ii} = \frac{1}{n} + (\mathbf{x}_i^* - \bar{\mathbf{x}})'(\tilde{X}'\tilde{X})^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$, where $\mathbf{x}_i^* = (x_{i1}, \ldots, x_{ip})'$ and $\tilde{X}$ is the centred $n \times p$ matrix of regressors. For $p = 1$, $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$.

Proof. (a) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p + 1$.

(b) The (i,i) entry of $H = HH'$ is $\sum_j h_{ij}^2$. Since it is at least h_{ii}^2, $h_{ii} \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$. For $j \neq i$, $h_{ij}^2 \le h_{ii} - h_{ii}^2 \le \frac{1}{4}$.

(c) $HX = X$ is direct; the intercept column gives $H\mathbf{1} = \mathbf{1}$. For $h_{ii} \ge 1/n$ see (d).

(d) With centred regressors $\tilde{X}$ orthogonal to $\mathbf{1}$, $H = \frac{1}{n}\mathbf{1}\mathbf{1}' + \tilde{X}(\tilde{X}'\tilde{X})^{-1}\tilde{X}'$; take diagonals (the second term is nonnegative). For $p = 1$, $\tilde{X}'\tilde{X} = S_{xx}$. ∎

As $\hat{y}_i = \sum_j h_{ij} y_j$, $h_{ii} = \partial \hat{y}_i / \partial y_i$ is the **leverage** of observation i; by (d) it grows with the Mahalanobis distance $M(i) = (\mathbf{x}_i^* - \bar{\mathbf{x}})'\{\tilde{X}'\tilde{X}/(n-1)\}^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$:

$$M(i) = (n-1)\Big(h_{ii} - \frac{1}{n}\Big).$$

The average leverage is $\bar{h} = (p+1)/n$, and a common rule flags $h_{ii} > 2\bar{h}$ (or $3\bar{h}$) as high leverage.

10.3 Residuals

Motivation. Scaling residuals by their standard errors, with the point deleted, gives t-comparable outlier statistics.

Theorem 10.2

$\mathbf{e} = (I-H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I-H))$. So $\operatorname{Var} e_i = \sigma^2(1-h_{ii})$, $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$, and $\operatorname{Corr}(e_i, e_j) = -h_{ij}/\sqrt{(1-h_{ii})(1-h_{jj})}$.

High-leverage points have small residual variance: the fit is pulled towards them. To compare residuals on a common scale:

Definition (Scaled residuals)

With $s^2 = MSE = \mathbf{e}'\mathbf{e}/(n-p-1)$ and $s^2_{(i)}$ the MSE computed without observation i:

- the **standardized** (internally studentized) residual is $r_i = \dfrac{e_i}{s\sqrt{1-h_{ii}}}$;
- the **studentized** (externally studentized) residual is $r_i^* = \dfrac{e_i}{s_{(i)}\sqrt{1-h_{ii}}}$.

r_i is not t: e_i and s are dependent (large $|e_i|$ inflates s). Deleting i removes this. Let $X_{(i)}$, $\mathbf{y}_{(i)}$ omit row i, with estimator $\hat\beta_{(i)}$ and prediction $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)}$.

Theorem 10.3 (Deletion formulas)

(a) $\hat\beta_{(i)} = \hat\beta - \dfrac{(X'X)^{-1}\mathbf{x}_i e_i}{1-h_{ii}}$.

(b) $y_i - \tilde y_i = \dfrac{e_i}{1-h_{ii}}$ (the **deleted** or PRESS residual).

(c) $(n-p-2)s^2_{(i)} = (n-p-1)s^2 - \dfrac{e_i^2}{1-h_{ii}}$.

Proof [Supplement]. (a) $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. With $A = X'X$, the Sherman–Morrison formula gives

$$(A - \mathbf{x}_i\mathbf{x}_i')^{-1} = A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i'A^{-1}}{1-h_{ii}}.$$

Multiplying by $X'\mathbf{y} - \mathbf{x}_i y_i$ and using $A^{-1}X'\mathbf{y} = \hat\beta$, $\mathbf{x}_i'\hat\beta = \hat y_i$, $\mathbf{x}_i'A^{-1}\mathbf{x}_i = h_{ii}$:

$$\hat\beta_{(i)} = \hat\beta - A^{-1}\mathbf{x}_i y_i + \frac{A^{-1}\mathbf{x}_i(\hat y_i - h_{ii}y_i)}{1-h_{ii}} = \hat\beta + \frac{A^{-1}\mathbf{x}_i\{\hat y_i - h_{ii}y_i - (1-h_{ii})y_i\}}{1-h_{ii}} = \hat\beta - \frac{A^{-1}\mathbf{x}_i e_i}{1-h_{ii}}.$$

(b) $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)} = \hat y_i - h_{ii}e_i/(1-h_{ii})$, so $y_i - \tilde y_i = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$.

(c) For $j \neq i$, the deleted residual is $y_j - \mathbf{x}_j'\hat\beta_{(i)} = e_j + h_{ji}e_i/(1-h_{ii})$. Summing squares over $j \neq i$ and using $\sum_j e_j h_{ji} = 0$ (since $H\mathbf{e} = \mathbf{0}$) and $\sum_j h_{ji}^2 = h_{ii}$:

$$SSE_{(i)} = \sum_{j\neq i} e_j^2 + \frac{2e_i}{1-h_{ii}}\sum_{j\neq i} h_{ji}e_j + \frac{e_i^2}{(1-h_{ii})^2}\sum_{j\neq i} h_{ji}^2.$$

Here $\sum_{j\neq i} e_j^2 = SSE - e_i^2$, $\sum_{j\neq i} h_{ji}e_j = -h_{ii}e_i$, and $\sum_{j\neq i} h_{ji}^2 = h_{ii} - h_{ii}^2$. So

$$SSE_{(i)} = SSE - e_i^2 - \frac{2h_{ii}e_i^2}{1-h_{ii}} + \frac{h_{ii}e_i^2}{1-h_{ii}} = SSE - \frac{e_i^2}{1-h_{ii}}.$$

■

Theorem 10.4 (Distribution of the studentized residual)

(a) $r_i^* = \dfrac{y_i - \tilde{y}_i}{s_{(i)}\sqrt{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i}} \sim t(n-p-2)$.

(b) $r_i^* = r_i\sqrt{\dfrac{n-p-2}{n-p-1-r_i^2}}$, a monotone function of r_i.

Proof. (a) y_i is independent of $\hat{\beta}_{(i)}$ and $s_{(i)}^2$, which are computed from the other $n-1$ observations; so $y_i - \tilde{y}_i \sim N(0, \sigma^2\{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i\})$, independent of $(n-p-2)s_{(i)}^2/\sigma^2 \sim \chi^2(n-p-2)$. By Sherman–Morrison, $1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i = 1 + h_{ii} + h_{ii}^2/(1-h_{ii}) = 1/(1-h_{ii})$, and with Theorem 10.3(b) the ratio equals $e_i/\{s_{(i)}\sqrt{1-h_{ii}}\}$.

(b) Substitute Theorem 10.3(c), $s_{(i)}^2 = s^2(n-p-1-r_i^2)/(n-p-2)$. ■

So an observation is flagged as an outlier when $|r_i^*| > t_{\alpha/2}(n-p-2)$. Since n tests are performed, a Bonferroni level α/n is safer.

10.4 Influence measures

An observation is influential if deleting it changes the fitted model a lot. The measures below combine the residual and the leverage.

Theorem 10.5 (Influence measures)

(a) **DFFITS**: $\mathrm{DFFITS}_{(i)} = \dfrac{\hat{y}_i - \tilde{y}_{i(i)}}{s_{(i)}\sqrt{h_{ii}}} = \left(\dfrac{h_{ii}}{1-h_{ii}}\right)^{1/2} r_i^*$, where $\tilde{y}_{i(i)} = \mathbf{x}_i'\hat{\beta}_{(i)}$.

(b) **Cook's distance**: $D_{(i)} = \dfrac{(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)})}{(p+1)s^2} = \dfrac{r_i^2}{p+1}\cdot\dfrac{h_{ii}}{1-h_{ii}}$.

(c) **COVRATIO**: $\mathrm{COVRATIO}_{(i)} = \dfrac{\det\{s_{(i)}^2(X_{(i)}'X_{(i)})^{-1}\}}{\det\{s^2(X'X)^{-1}\}} = \dfrac{1}{(1-h_{ii})}\left(\dfrac{n-p-2+r_i^{*2}}{n-p-1}\right)^{-(p+1)}$.

(d) **Andrews–Pregibon**: with $Z = (X, \mathbf{y})$, $\mathrm{AP}_{(i)} = \dfrac{\det(Z_{(i)}'Z_{(i)})}{\det(Z'Z)} = 1 - h_{ii} - \dfrac{e_i^2}{SSE}$.

Proof [Supplement]. (a) By Theorem 10.3(a), $\hat{y}_i - \tilde{y}_{i(i)} = \mathbf{x}_i'(\hat{\beta} - \hat{\beta}_{(i)}) = h_{ii}e_i/(1-h_{ii})$; divide by $s_{(i)}\sqrt{h_{ii}}$.

(b) $(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)}) = \frac{e_i^2}{(1-h_{ii})^2}\mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i = \frac{h_{ii}e_i^2}{(1-h_{ii})^2}$; divide by $(p+1)s^2$.

(c) By the matrix determinant lemma,

$$\det(X_{(i)}'X_{(i)}) = \det(X'X)\,(1-h_{ii}),$$

so the ratio is $(s^2_{(i)}/s^2)^{p+1}/(1-h_{ii})$. By Theorem 10.3(c), $s^2_{(i)}/s^2 = (n-p-1-r_i^2)/(n-p-2)$, and by Theorem 10.4(b), $n-p-2+r_i^{*2} = (n-p-2)(n-p-1)/(n-p-1-r_i^2)$. Hence $s^2_{(i)}/s^2 = (n-p-1)/(n-p-2+r_i^{*2})$, which gives the formula.

(d) By the determinant lemma, $\det(Z'_{(i)}Z_{(i)})/\det(Z'Z) = 1-\mathbf{z}_i'(Z'Z)^{-1}\mathbf{z}_i$, $\mathbf{z}_i' = (\mathbf{x}_i', y_i)$: the leverage in the regression on Z, which equals $h_{ii} + e_i^2/SSE$ (partitioned inverse). ■

Rules of thumb (Belsley, Kuh and Welsch; Cook):

- $|\text{DFFITS}_{(i)}| > 2\sqrt{(p+1)/n}$;
- $D_{(i)} > F_{0.50}(p+1, n-p-1)$ (roughly 1): deleting i moves $\hat\beta$ outside the 50% confidence ellipsoid of Theorem 8.2;
- $|\text{COVRATIO}_{(i)} - 1| > 3(p+1)/n$;
- small $\text{AP}_{(i)}$.

Every measure grows with both $|r_i|$ and h_{ii}: influence is "outlyingness × leverage".

Example 10.1 (Two kinds of unusual point)

For $n = 21$ children (age at first word x, aptitude y), $\hat y = 109.87 - 1.13x$. Observation 18 ($x = 42$) has $h_{18,18} = 0.65 > 3\bar h = 0.29$ and a small residual ($r^* = -0.85$), yet the largest $D_{(18)} = 0.68$ and $\text{DFFITS}_{(18)} = -1.16$: influential. Observation 19 has $r^*_{19} = 3.61 > t_{0.025}(18) = 2.10$: an outlier, but with ordinary leverage ($h = 0.053$), so Cook's distance is 0.22. Check the outlier's measurement; assess dependence on the high-leverage point.

What to do. Check unusual points for recording errors; a genuine one may signal a wrong model. Robust regression or fits with and without the point beat silent deletion. Single-deletion diagnostics can miss groups of points that mask each other.

10.5 Selected exercises

Motivation. A point on the fitted surface, leverage one, and why standardized residuals are not t.

Exercise 10.1 (Deleting a point on the fitted surface)

Suppose $y_i = \hat y_i$ and $X_{(i)}$ has full column rank. Show from the normal equations that deleting i leaves the LSE unchanged, and find $s^2_{(i)}$ in terms of s^2.

Solution.

Step 1. $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. Hence, for the full-data estimator $\hat\beta$,

$$X'_{(i)}(\mathbf{y}_{(i)} - X_{(i)}\hat\beta) = X'(\mathbf{y} - X\hat\beta) - \mathbf{x}_i(y_i - \mathbf{x}_i'\hat\beta) = \mathbf{0} - \mathbf{x}_i e_i = \mathbf{0},$$

since $X'\mathbf{e} = \mathbf{0}$ and $e_i = 0$.

Step 2. So $\hat\beta$ solves the normal equations of the reduced data. As $X'_{(i)}X_{(i)}$ is nonsingular, the solution is unique: $\hat\beta_{(i)} = \hat\beta$. (This is the case $e_i = 0$ of Theorem 10.3(a).)

Step 3. The remaining residuals are unchanged, so $SSE_{(i)} = \sum_{j\neq i} e_j^2 = SSE$, now with $n-p-2$ degrees of freedom: $s^2_{(i)} = \frac{n-p-1}{n-p-2}s^2$, which is larger than s^2 when $SSE > 0$.

Such a point has zero Cook's distance however large its leverage, illustrating that leverage alone does not make a point influential. ■

Exercise 10.2 (Leverage equal to one)

Suppose $h_{ii} = 1$.
(a) Show that $h_{ij} = 0$ for $j \neq i$, and that $\hat{y}_i = y_i$ and $e_i = 0$ for every response vector $\mathbf{y}$.
(b) Show that $X'_{(i)}X_{(i)}$ is singular, so that β cannot be estimated without observation i; conversely, if $X_{(i)}$ has full column rank, then $h_{ii} < 1$.
(c) In simple regression with $x = (0, \dots, 0, 1)$ (with $n \geq 2$), compute h_{nn}.

Solution [Supplement].

(a) By Theorem 10.1(b), $h_{ii} = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$, i.e. $1 = 1 + \sum_{j \neq i} h_{ij}^2$, so every $h_{ij} = 0$ $(j \neq i)$. Then $\hat{y}_i = \sum_j h_{ij} y_j = y_i$ and $e_i = 0$, whatever $\mathbf{y}$ is.

(b) By the matrix determinant lemma (as in the proof of Theorem 10.5(c)),

$$\det(X'_{(i)}X_{(i)}) = \det(X'X - \mathbf{x}_i\mathbf{x}_i') = \det(X'X)\,\{1 - \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i\} = \det(X'X)(1 - h_{ii}).$$

So $h_{ii} = 1$ makes $X'_{(i)}X_{(i)}$ singular, i.e. $X_{(i)}$ rank deficient. Conversely, if $X_{(i)}$ has full column rank, the determinant is positive, so $h_{ii} \neq 1$, and $h_{ii} < 1$ by Theorem 10.1(b).

(c) $\bar{x} = 1/n$ and $S_{xx} = (n-1)/n^2 + (1 - 1/n)^2 = (n-1)/n$. By Theorem 10.1(d),

$$h_{nn} = \frac{1}{n} + \frac{(1 - 1/n)^2}{(n-1)/n} = \frac{1}{n} + \frac{n-1}{n} = 1.$$

The others all have $x = 0$, so the point at $x = 1$ alone determines the slope, and the line passes through it. ■

Exercise 10.3 (Standardized residuals are bounded)

Assume $h_{ii} < 1$.
(a) Show that $r_i^2 \leq n - p - 1$. Conclude that r_i does not have a t distribution.
(b) Show that the map $r_i \mapsto r_i^*$ of Theorem 10.4(b) is a strictly increasing bijection from $(-\sqrt{n-p-1}, \sqrt{n-p-1})$ onto $\mathbb{R}$.

Solution [Supplement].

(a) By Theorem 10.3(c), $0 \leq (n-p-2)s_{(i)}^2 = (n-p-1)s^2 - e_i^2/(1-h_{ii})$. Hence $e_i^2/\{s^2(1-h_{ii})\} \leq n-p-1$, i.e. $r_i^2 \leq n-p-1$, with equality iff $SSE_{(i)} = 0$. A t variable takes arbitrarily large values with positive probability, so r_i cannot be t-distributed.

(b) On $|r| < \sqrt{n-p-1}$ let $f(r) = r\sqrt{(n-p-2)/(n-p-1-r^2)}$, continuous and odd. For $0 \leq r_1 < r_2$ both positive factors increase, so $f(r_1) < f(r_2)$; by oddness f is strictly increasing, with $f \to \pm\infty$ at $\pm\sqrt{n-p-1}$, hence onto $\mathbb{R}$ (IVT). So r_i^* and r_i rank alike, but only r_i^* has the unbounded $t(n-p-2)$ law. ■

Chapter 11. Variable Selection

With many candidate regressors we want a **parsimonious** model: enough variables to explain and predict y, no more, since each adds variance.

11.1 Consequences of omitting variables

Partition the **full model** as

$$\mathbf{y} = X\beta + \epsilon = X_k\beta_k + X_r\beta_r + \epsilon, \qquad \epsilon \sim N(\mathbf{0}, \sigma^2 I),$$

where X_k $(n \times (k+1))$ contains the intercept and k regressors and X_r $(n \times r,\ r = p-k)$ the others. The full-model estimator $\hat\beta = (\hat\beta_k', \hat\beta_r')'$ and $\hat\sigma^2 = SSE/(n-p-1)$ are compared with the **reduced model** estimators

$$\tilde\beta_k = (X_k'X_k)^{-1}X_k'\mathbf{y}, \qquad \tilde\sigma^2 = \frac{\mathbf{y}'\{I - X_k(X_k'X_k)^{-1}X_k'\}\mathbf{y}}{n-k-1}.$$

Theorem 11.1

Let $A = (X_k'X_k)^{-1}X_k'X_r$ (the **alias matrix**) and $H_k = X_k(X_k'X_k)^{-1}X_k'$. If the full model is true:

(a) $E\tilde\beta_k = \beta_k + A\beta_r$ and $\operatorname{Var}\tilde\beta_k = \sigma^2(X_k'X_k)^{-1}$; so $\operatorname{MSE}(\tilde\beta_k) = E(\tilde\beta_k - \beta_k)(\tilde\beta_k - \beta_k)' = \sigma^2(X_k'X_k)^{-1} + A\beta_r\beta_r'A'$.

(b) $E\tilde\sigma^2 = \sigma^2 + \dfrac{\beta_r'X_r'(I-H_k)X_r\beta_r}{n-k-1} \geq \sigma^2$.

(c) For predicting $E(y) = \mathbf{x}'\beta = \mathbf{x}_k'\beta_k + \mathbf{x}_r'\beta_r$ by $\tilde y = \mathbf{x}_k'\tilde\beta_k$:

$$\operatorname{MSE}(\tilde y) = \sigma^2\mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + (\mathbf{x}_k'A\beta_r - \mathbf{x}_r'\beta_r)^2.$$

Proof. (a) $E\tilde\beta_k = (X_k'X_k)^{-1}X_k'(X_k\beta_k + X_r\beta_r) = \beta_k + A\beta_r$; the variance is as usual; MSE = Var + bias · bias′.

(b) Theorem 5.4(a) with $I - H_k$, whose trace is $n-k-1$ and which annihilates $X_k\beta_k$.

(c) MSE = Var + bias^2 with bias $\mathbf{x}_k'(\beta_k + A\beta_r) - \mathbf{x}'\beta$. ■

So omitting relevant variables biases the remaining coefficients (unless $X_k'X_r = 0$) and inflates the residual variance estimate. In exchange the variance decreases:

Theorem 11.2

(a) $\operatorname{Var}(\hat\beta_k) - \operatorname{Var}(\tilde\beta_k)$ is nonnegative definite.

(b) $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{Var}(\mathbf{x}_k'\tilde\beta_k)$.

(c) If $\operatorname{Var}(\hat\beta_r) - \beta_r\beta_r'$ is nonnegative definite, then $\operatorname{Var}(\hat\beta_k) - \operatorname{MSE}(\tilde\beta_k)$ is nonnegative definite and $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{MSE}(\mathbf{x}_k'\tilde\beta_k)$.

Proof. By the partitioned-inverse formula, with $B = X_r'(I-H_k)X_r$ (nonsingular),

$$\sigma^{-2}\operatorname{Var}(\hat\beta) = (X'X)^{-1} = \begin{pmatrix} (X_k'X_k)^{-1} + AB^{-1}A' & -AB^{-1} \\ -B^{-1}A' & B^{-1} \end{pmatrix}.$$

(a) $\operatorname{Var}\hat\beta_k - \operatorname{Var}\tilde\beta_k = \sigma^2 AB^{-1}A' \geq 0$.

(b) With $\mathbf{x} = (\mathbf{x}_k', \mathbf{x}_r')'$ and $\mathbf{u} = A'\mathbf{x}_k - \mathbf{x}_r$,

$$\sigma^{-2}\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + \mathbf{u}'B^{-1}\mathbf{u} \geq \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k.$$

(c) $\operatorname{Var}\hat{\beta}_r = \sigma^2 B^{-1}$, so the assumption reads $\sigma^2 B^{-1} \geq \beta_r\beta_r'$. Then $\operatorname{Var}\hat{\beta}_k - \operatorname{MSE}(\tilde{\beta}_k) = A(\sigma^2 B^{-1} - \beta_r\beta_r')A' \geq 0$, and similarly $\operatorname{Var}(\mathbf{x}'\hat{\beta}) - \operatorname{MSE}(\mathbf{x}_k'\tilde{\beta}_k) = \mathbf{u}'(\sigma^2 B^{-1} - \beta_r\beta_r')\mathbf{u} \geq 0$. ■

So dropping variables with coefficients small relative to their SEs ($\beta_r\beta_r' \leq \operatorname{Var}\hat{\beta}_r$) reduces MSE despite bias; if $\beta_r = \mathbf{0}$, both are unbiased and the reduced model has smaller variance.

11.2 Search methods

With p candidate regressors there are 2^p subsets (each variable in or out), so exhaustive search becomes infeasible for large p ($2^{50} \approx 10^{15}$).

All possible regressions (best subset). Fit all 2^p models; for each size k keep the best by R_k^2 (equivalently SSE_k, MSE_k), then compare sizes with a penalized criterion (Section 11.3).

Backward elimination.

(1) Fit the full model.

(2) Compute the partial F statistic of each variable in the model.

(3) Let F_L be the smallest, and F_C the critical value at level α_{drop}.

(4) If $F_L < F_C$, delete that variable, refit and return to (2); otherwise stop.

This needs at most $p(p+1)/2$ fits.

Forward selection. Start empty; add the variable with the largest partial F if it exceeds the level-α_{add} critical value; else stop.

Stepwise selection. Forward selection, re-examining included variables after each addition and removing those below α_{drop} ($\alpha_{\text{drop}} \geq \alpha_{\text{add}}$ to avoid cycling).

The methods may disagree, none guarantees the best subset, and the levels are not true significance levels.

Example 11.1

Hald's cement data ($n = 13$, ingredients $x_1, \ldots, x_4$, $x_1 + x_2 + x_3 + x_4 \approx 100$, x_2, x_4 highly correlated): backward elimination ($\alpha = 0.05$) removes x_3 (partial $F = 0.02$), then x_4 ($1.86 < F_{0.05}(1, 9) = 5.12$), ending at $\{x_1, x_2\}$ ($R^2 = 0.979$). Forward selection enters x_4 ($R^2 = 0.675$), then x_1, and stops (x_2: $5.03 < 5.12$) at $\{x_1, x_4\}$ ($R^2 = 0.972$). Stepwise ($\alpha_{\text{add}} = 0.10$) enters x_2, removes x_4, ending at $\{x_1, x_2\}$: nearly equivalent information gives different, almost equally good models.

11.3 Selection criteria

Motivation. R^2 cannot choose a size; MSE, adjusted R^2 and Mallows' C_p penalize complexity.

For a model with k regressors (plus intercept):

(1) residual mean square $MSE_k = SSE_k/(n-k-1)$;

(2) $R_k^2 = 1 - SSE_k/SST$;

(3) adjusted R^2: $R_{a,k}^2 = 1 - \dfrac{n-1}{n-k-1}(1-R_k^2) = 1 - \dfrac{MSE_k}{SST/(n-1)}$;

(4) Mallows' $C_k = \dfrac{SSE_k}{\hat\sigma^2} + 2(k+1) - n$, with $\hat\sigma^2$ the MSE of the full model.

R_k^2 never decreases, so it compares only equal sizes. MSE_k can increase with a variable (a lost d.f. may outweigh the drop in SSE), and maximizing $R_{a,k}^2$ is minimizing MSE_k.

Mallows' C_p. We judge the reduced model by the total standardized mean squared error of its fitted values at the n design points:

$$\Gamma_k = \frac{1}{\sigma^2}\sum_{i=1}^{n} \mathrm{MSE}(\tilde y_i), \qquad \tilde y_i = \mathbf{x}_{ki}'\tilde\beta_k.$$

Theorem 11.3

$$\Gamma_k = (k+1) + \frac{\beta_r' X_r'(I-H_k)X_r\beta_r}{\sigma^2} = \frac{E(SSE_k)}{\sigma^2} + 2(k+1) - n.$$

Hence C_k estimates Γ_k, and if the reduced model is correct ($\beta_r = \mathbf{0}$), $E(SSE_k) = (n-k-1)\sigma^2$ and $C_k \approx k+1$.

Proof [Supplement]. By Theorem 11.1(c), the variance part sums to

$$\sum_i \mathbf{x}_{ki}'(X_k'X_k)^{-1}\mathbf{x}_{ki} = \mathrm{tr}\Big\{(X_k'X_k)^{-1}\sum_i \mathbf{x}_{ki}\mathbf{x}_{ki}'\Big\} = \mathrm{tr}\{(X_k'X_k)^{-1}X_k'X_k\} = k+1.$$

The bias vector over the design points is $X_kA\beta_r - X_r\beta_r = -(I-H_k)X_r\beta_r$, with squared length $\beta_r'X_r'(I-H_k)X_r\beta_r$. By Theorem 11.1(b), this equals $E(SSE_k) - (n-k-1)\sigma^2$. Substituting gives the second expression. ■

So we look for models with small C_k and C_k close to $k+1$; a C_k well above $k+1$ signals bias from omitted variables. For the full model, $C_p = p+1$ identically.

11.4 Selected exercises

Motivation. Omitted-variable bias for a line, and adjusted R^2 and C_k via the partial F.

Exercise 11.1 (Omitted-variable bias for a straight line)

The straight line $\hat y = \hat\beta_0 + \hat\beta_1 x_1$ is fitted by least squares, but the true mean is $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ (errors uncorrelated with variance σ^2). Write $S_{11} = \sum(x_{i1}-\bar x_1)^2 > 0$, $S_{22} = \sum(x_{i2}-\bar x_2)^2$ and $S_{12} = \sum(x_{i1}-\bar x_1)(x_{i2}-\bar x_2)$.
(a) Is $\hat\beta_1$ biased? Find $E\hat\beta_1$.
(b) Find the expectation of the residual mean square of the fitted line.

Solution.

(a) With $k_i = (x_{i1}-\bar x_1)/S_{11}$, $\hat\beta_1 = \sum k_i y_i$, $\sum k_i = 0$ and $\sum k_i x_{i1} = 1$ (Section 4.1). Hence

$$E\hat\beta_1 = \sum k_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) = \beta_1 + \beta_2\sum k_i x_{i2} = \beta_1 + \beta_2\frac{S_{12}}{S_{11}}.$$

So $\hat{\beta}_1$ is biased unless $\beta_2 = 0$ or $S_{12} = 0$. S_{12}/S_{11}, the slope of x_2 on x_1, is the alias-matrix entry of Theorem 11.1: $\hat{\beta}_1$ absorbs the part of x_2's effect travelling with x_1.

(b) By Theorem 11.1(b) with $k = 1$ and $X_r = \mathbf{x}_2$, the excess is $\beta_2^2\, \mathbf{x}_2'(I - H_k)\mathbf{x}_2/(n-2)$, where $\mathbf{x}_2'(I - H_k)\mathbf{x}_2$ is the residual sum of squares of the regression of x_2 on $(1, x_1)$, namely $S_{22} - S_{12}^2/S_{11}$ (Theorem 3.3). So

$$E(MSE) = \sigma^2 + \frac{\beta_2^2(S_{22} - S_{12}^2/S_{11})}{n-2} \geq \sigma^2.$$

■

Exercise 11.2 (Adjusted R^2 and the partial F statistic)

A model with k regressors (and intercept) is enlarged by r more; F is their partial F with the larger model's residual mean square. With $SSE_{k+r} > 0$, show that adjusted R^2 increases iff $F > 1$.

Solution [Supplement].

Step 1. Since $R_a^2 = 1 - MSE/\{SST/(n-1)\}$ with SST common to both models, R_a^2 increases iff $MSE_{k+r} < MSE_k$.

Step 2. By definition $F = \{(SSE_k - SSE_{k+r})/r\}/MSE_{k+r}$, so $SSE_k = SSE_{k+r} + rF\, MSE_{k+r} = (n - k - r - 1 + rF)MSE_{k+r}$, and

$$MSE_k = \frac{SSE_k}{n-k-1} = \frac{n-k-r-1+rF}{n-k-1}\, MSE_{k+r}.$$

Step 3. With $MSE_{k+r} > 0$, $MSE_k > MSE_{k+r}$ iff $n - k - r - 1 + rF > n - k - 1$ iff $F > 1$. So adjusted R^2 admits a variable far more easily than a significance test, which would require $F > F_\alpha$ (about 4 for one variable at $\alpha = 0.05$). ■

Exercise 11.3 (Mallows' C_k and the partial F statistic)

Let F_0 be the partial F statistic for dropping the $r = p - k$ regressors of X_r from the full model. Show that

$$C_k = (k+1) + (p-k)(F_0 - 1),$$

so that $C_k \leq k + 1$ if and only if $F_0 \leq 1$.

Solution [Supplement]. Let $\hat{\sigma}^2 = SSE/(n-p-1)$ be the full-model residual mean square. By definition of F_0, $SSE_k = SSE + (p-k)F_0\hat{\sigma}^2 = \{(n-p-1) + (p-k)F_0\}\hat{\sigma}^2$. Substituting into $C_k = SSE_k/\hat{\sigma}^2 + 2(k+1) - n$,

$$C_k = (n-p-1) + (p-k)F_0 + 2(k+1) - n = (k+1) + (p-k)F_0 - (p-k).$$

Since $F_0 \approx 1$ on average when $\beta_r = \mathbf{0}$ (its mean is $\nu/(\nu-2)$ with $\nu = n-p-1$), this matches $C_k \approx k + 1$ in Theorem 11.3. ■

Chapter 12. Generalized Least Squares

12.1 The general linear model

Uncorrelated equal-variance errors fail for time series, clustered or heteroscedastic data. The **general linear regression model** is

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 V),$$

with V a known positive definite matrix, X of size $n \times (p+1)$ and full column rank. Ordinary least squares (OLS) is no longer the BLUE.

Since V is symmetric positive definite, $V = P\Lambda P'$ with P orthogonal, and $T = P\Lambda^{1/2}$ satisfies $V = TT'$. Multiply the model by T^{-1}:

$$\mathbf{z} = W\beta + \eta, \qquad \mathbf{z} = T^{-1}\mathbf{y}, \quad W = T^{-1}X, \quad \eta = T^{-1}\epsilon,$$

and $\operatorname{Var}\eta = \sigma^2 T^{-1}V(T^{-1})' = \sigma^2 T^{-1}TT'(T')^{-1} = \sigma^2 I_n$. The transformed model satisfies the usual assumptions.

Theorem 12.1 (Generalized least squares)

The BLUE of β is the **generalized least squares** (GLS) estimator

$$\hat\beta_G = (W'W)^{-1}W'\mathbf{z} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y},$$

which minimizes $(\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. It satisfies $E\hat\beta_G = \beta$ and $\operatorname{Var}\hat\beta_G = \sigma^2(X'V^{-1}X)^{-1}$.

Proof. Apply Gauss–Markov (Theorem 7.1) to the transformed model (T invertible, so linear unbiased estimators correspond). With $(T^{-1})'T^{-1} = V^{-1}$: $W'W = X'V^{-1}X$, $W'\mathbf{z} = X'V^{-1}\mathbf{y}$, $\|\mathbf{z} - W\beta\|^2 = (\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. Then $E\hat\beta_G = \beta$ and $\operatorname{Var}\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}(\sigma^2 V)V^{-1}X(X'V^{-1}X)^{-1} = \sigma^2(X'V^{-1}X)^{-1}$. ∎

When V is diagonal, $V = \operatorname{diag}(v_i)$, GLS is **weighted least squares** with weights $w_i = 1/v_i$: minimize $\sum w_i(y_i - \mathbf{x}_i'\beta)^2$, giving less weight to noisier observations.

12.2 Properties

OLS under correlated errors. The OLS estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is still unbiased, but

$$\operatorname{Var}\hat\beta = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1},$$

which is at least $\sigma^2(X'V^{-1}X)^{-1}$ (by Gauss–Markov), with equality when $V = I$ (more generally, when the column space of X is invariant under V).

Residuals and σ^2. The GLS residual vector is $\mathbf{e}_G = \mathbf{y} - X\hat\beta_G = \{I - X(X'V^{-1}X)^{-1}X'V^{-1}\}\mathbf{y}$, with

$$E\mathbf{e}_G = \mathbf{0}, \qquad \operatorname{Var}\mathbf{e}_G = \sigma^2\{V - X(X'V^{-1}X)^{-1}X'\}.$$

Theorem 12.2

$SSE(\hat\beta_G) = \mathbf{e}_G' V^{-1}\mathbf{e}_G$ satisfies $E\,SSE(\hat\beta_G) = (n-p-1)\sigma^2$, so

$$\tilde\sigma^2 = \frac{(\mathbf{y} - X\hat\beta_G)' V^{-1}(\mathbf{y} - X\hat\beta_G)}{n-p-1}$$

is unbiased for σ^2, and $\widehat{\mathrm{Var}}(\hat\beta_G) = \tilde\sigma^2 (X'V^{-1}X)^{-1}$.

Proof. $SSE(\hat\beta_G)$ is the residual sum of squares of the transformed model, $\|\mathbf{z} - W\hat\beta_G\|^2$, whose expectation is $(n-p-1)\sigma^2$ (Theorem 6.2). Directly: by Theorem 5.4(a),

$$\begin{aligned} E(\mathbf{e}_G' V^{-1}\mathbf{e}_G) &= \sigma^2 \mathrm{tr}[V^{-1}\{V - X(X'V^{-1}X)^{-1}X'\}] \\ &= \sigma^2\{n - \mathrm{tr}((X'V^{-1}X)^{-1}X'V^{-1}X)\} = \sigma^2(n-p-1). \end{aligned}$$

■

By contrast, the OLS estimate $\hat\sigma^2 = \mathbf{y}'(I-H)\mathbf{y}/(n-p-1)$ has

$$E\hat\sigma^2 = \frac{\sigma^2}{n-p-1}[\mathrm{tr}V - \mathrm{tr}\{(X'X)^{-1}X'VX\}],$$

which in general differs from σ^2, so the usual OLS standard errors $\hat\sigma^2(X'X)^{-1}$ are biased in two ways under correlated errors.

Under normality, all the inference of Chapters 6–8 carries over to the transformed model: for instance $\hat\beta_G \sim N(\beta, \sigma^2(X'V^{-1}X)^{-1})$, $(n-p-1)\tilde\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and t and F tests use $\tilde\sigma^2$.

Example 12.1 (Averages with unequal precision)

Suppose $y_i = \mu + \epsilon_i$ where y_i is the average of m_i measurements, so $\mathrm{Var}\,\epsilon_i = \sigma^2/m_i$ and $V = \mathrm{diag}(1/m_i)$. Then $X = \mathbf{1}$ and

$$\hat\mu_G = \frac{\sum m_i y_i}{\sum m_i}, \qquad \mathrm{Var}\,\hat\mu_G = \frac{\sigma^2}{\sum m_i},$$

the grand average of all measurements. The OLS estimate $\bar y$ has variance $\frac{\sigma^2}{n^2}\sum \frac{1}{m_i} \geq \frac{\sigma^2}{\sum m_i}$ by the harmonic–arithmetic mean inequality.

12.3 Selected exercises

Motivation. OLS facts under correlated errors (Section 12.2), and WLS through the origin.

Exercise 12.1 (OLS under a general covariance)

Let $\mathbf{y} = X\beta + \epsilon$ with $E\epsilon = \mathbf{0}$, $\mathrm{Var}\,\epsilon = \sigma^2 V$, V known positive definite and X of full column rank $p+1$. Let $\tilde\beta = (X'X)^{-1}X'\mathbf{y}$ and $\hat\sigma^2 = \mathbf{y}'(I-H)\mathbf{y}/(n-p-1)$.
(a) Derive $E\tilde\beta$ and $\mathrm{Var}\,\tilde\beta$.
(b) Derive $E\hat\sigma^2$.
(c) Show that if $VX = XQ$ for some matrix Q, then $\tilde\beta = \hat\beta_G$.
(d) Let $V = (1-\rho)I_n + \rho J_n$ with $-1/(n-1) < \rho < 1$ (equicorrelated errors), and let X contain the intercept column. Show that OLS and GLS coincide.

Solution.

(a) $E\tilde{\beta} = (X'X)^{-1}X'X\beta = \beta$ and

$$\operatorname{Var}\tilde{\beta} = (X'X)^{-1}X'(\sigma^2 V)X(X'X)^{-1} = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1}.$$

(b) By Theorem 5.4(a) with mean $X\beta$, covariance $\sigma^2 V$ and $(I-H)X = 0$,

$$\begin{aligned} E\{\mathbf{y}'(I-H)\mathbf{y}\} &= \sigma^2 \operatorname{tr}\{(I-H)V\} = \sigma^2\big[\operatorname{tr}V - \operatorname{tr}\{X(X'X)^{-1}X'V\}\big] \\ &= \sigma^2\big[\operatorname{tr}V - \operatorname{tr}\{(X'X)^{-1}X'VX\}\big], \end{aligned}$$

using $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Dividing by $n-p-1$ gives the formula of Section 12.2.

(c) VX has full column rank (V is nonsingular), so XQ does, and the square matrix Q is nonsingular. Then $V^{-1}X = XQ^{-1}$ and $X'V^{-1} = (Q^{-1})'X'$, so

$$\hat{\beta}_G = \{(Q^{-1})'X'X\}^{-1}(Q^{-1})'X'\mathbf{y} = (X'X)^{-1}Q'(Q^{-1})'X'\mathbf{y} = (X'X)^{-1}X'\mathbf{y} = \tilde{\beta}.$$

(d) V has eigenvalue $1+(n-1)\rho$ on $\mathbf{1}$ and $1-\rho$ on $\mathbf{1}^{\perp}$, both positive, so V is positive definite. If the intercept is the first column, $\mathbf{1} = X\mathbf{e}_1$, so $J_n X = \mathbf{1}\mathbf{1}'X = X\mathbf{e}_1\mathbf{1}'X$ and

$$VX = (1-\rho)X + \rho X\mathbf{e}_1\mathbf{1}'X = X\{(1-\rho)I + \rho\,\mathbf{e}_1\mathbf{1}'X\} = XQ.$$

By (c), $\tilde{\beta} = \hat{\beta}_G$: with equicorrelated errors and an intercept, OLS is BLUE, although its usual standard errors and $\hat{\sigma}^2$ are still biased by (a)–(b). ■

Exercise 12.2 (Weighted least squares through the origin)

Let $y_i = \beta x_i + \epsilon_i$, $i = 1, \dots, n$, with $x_i > 0$, uncorrelated errors, $E\epsilon_i = 0$ and $\operatorname{Var}\epsilon_i = \sigma^2 x_i$.
(a) Show that the GLS estimator is $\hat{\beta}_G = \bar{y}/\bar{x}$, with $\operatorname{Var}\hat{\beta}_G = \sigma^2/(n\bar{x})$.
(b) Find the variance of the OLS estimator $\tilde{\beta} = \sum x_i y_i / \sum x_i^2$.
(c) Verify directly that $\operatorname{Var}\tilde{\beta} \geq \operatorname{Var}\hat{\beta}_G$, with equality iff all x_i are equal.

Solution [Supplement].

(a) $V = \operatorname{diag}(x_i)$ and $X = (x_1, \dots, x_n)'$, so $X'V^{-1}X = \sum x_i^2/x_i = \sum x_i$ and $X'V^{-1}\mathbf{y} = \sum y_i$. By Theorem 12.1, $\hat{\beta}_G = \sum y_i / \sum x_i = \bar{y}/\bar{x}$ and $\operatorname{Var}\hat{\beta}_G = \sigma^2/\sum x_i = \sigma^2/(n\bar{x})$.

(b) $\operatorname{Var}\tilde{\beta} = \sum x_i^2 \operatorname{Var} y_i / (\sum x_i^2)^2 = \sigma^2 \sum x_i^3 / (\sum x_i^2)^2$.

(c) Need $(\sum x_i^2)^2 \leq \sum x_i \sum x_i^3$: Cauchy–Schwarz with $a_i = x_i^{1/2}$, $b_i = x_i^{3/2}$, equality iff $x_i = b_i/a_i$ is constant (cf. Section 12.2). The GLS estimator is the *ratio estimator*, natural when the spread of y grows with x. ■

Chapter 13. Indicator Variables

13.1 One indicator variable

Qualitative explanatory variables (sex, treatment, region) enter a regression through **indicator** (dummy) variables taking the values 0 and 1.

Parallel lines. Suppose y is the score on a test, x_1 the hours of preparation, and $x_2 = 1$ for students who attended a review session and 0 otherwise. The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

gives two response functions with a common slope:

$$E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + \beta_1 x_1, & x_2 = 1. \end{cases}$$

β_1 is the common effect of x_1, β_2 the vertical distance between the parallel lines; one model uses all n observations for the common slope and σ^2.

Interaction. If the effect of x_1 may differ between groups, add the product term:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \epsilon_i, \qquad E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + (\beta_1 + \beta_3) x_1, & x_2 = 1. \end{cases}$$

Now β_2 is the difference in intercepts and β_3 the difference in slopes. The hypotheses of Chapter 5 become ordinary tests in one model:

- parallel lines: $H_0 : \beta_3 = 0$ (a t test);
- identical lines: $H_0 : \beta_2 = \beta_3 = 0$ (a partial F test with 2 degrees of freedom).

These agree with the separate-fit tests of Section 5.3: the interaction model has the four parameters of the two lines and the same residual sum of squares (Exercise 13.4).

Example 13.1

20 firms: y time to adopt an innovation, x_1 size, $x_2 = 1$ for stock companies. $\hat{y} = 33.87 - 0.102 x_1 + 8.06 x_2$: larger firms adopt faster (about 1 month per 10 size units), stock companies about 8 months later at every size. The interaction has $\hat{\beta}_3 = -0.0004$ (SE 0.018, $t_0 = -0.02$): parallel lines suffice.

13.2 Several categories

A qualitative variable with q categories needs $q - 1$ indicators. With three categories E_1, E_2, E_3, define

$$x_2 = \mathbf{1}\{E_1\}, \qquad x_3 = \mathbf{1}\{E_2\},$$

with E_3 the **baseline**:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i, \qquad E(y) = \begin{cases} (\beta_0 + \beta_2) + \beta_1 x_1, & E_1, \\ (\beta_0 + \beta_3) + \beta_1 x_1, & E_2, \\ \beta_0 + \beta_1 x_1, & E_3. \end{cases}$$

So β_2 and β_3 compare E_1 and E_2 with the baseline, and $\beta_2 - \beta_3$ compares E_1 with E_2.

The dummy variable trap. q indicators plus an intercept are linearly dependent (they sum to $\mathbf{1}$), so $X'X$ is singular and β unidentifiable. Drop one indicator (baseline coding), drop the intercept, or constrain e.g. $\sum(\text{effects})= 0$.

Interactions and several factors. Separate slopes are obtained with the products x_1x_2 and x_1x_3:

$$E(y) = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_1x_2 + \beta_5x_1x_3,$$

so in E_1 the line is $(\beta_0 + \beta_2) + (\beta_1 + \beta_4)x_1$, and so on. Each qualitative factor gets its indicators, interactions are products of indicators, and an indicator-only model is ANOVA written as regression.

13.3 Piecewise linear regression

Sometimes the slope changes at a known point x_w (a threshold or a policy change), while the response stays continuous. With $x_2 = \mathbf{1}\{x_1 > x_w\}$,

$$y_i = \beta_0 + \beta_1x_{i1} + \beta_2(x_{i1} - x_w)x_{i2} + \epsilon_i,$$

so that

$$E(y) = \begin{cases} \beta_0 + \beta_1x_1, & x_1 \le x_w, \\ (\beta_0 - \beta_2x_w) + (\beta_1 + \beta_2)x_1, & x_1 > x_w. \end{cases}$$

The lines meet at x_w; β_2 is the slope change, and $H_0 : \beta_2 = 0$ tests for a break. $(x_1 - x_w)_+ = (x_1 - x_w)x_2$ is a **truncated power basis** function; each further knot adds one (a linear spline), and adding x_2 allows a jump.

Example 13.2

Unit cost y against lot size x_1 for eight lots, with a suspected change at $x_w = 500$:

x_1	300	340	400	480	570	650	720	800
y	475	440	452	377	355	257	249	139

The fit is $\hat{y} = 589.5 - 0.395x_1 - 0.389(x_1 - 500)_+$: the cost falls by 0.40 per unit up to 500 and by $0.40 + 0.39 = 0.78$ per unit beyond.

13.4 Selected exercises

Motivation. Indicator models are ordinary regressions: a jump test, the two-sample t test, the pooled within-group slope, and interaction vs two separate fits.

Exercise 13.1 (Testing for a jump)

Consider

$$y_i = \beta_0 + \beta_1x_{i1} + \beta_2(x_{i1} - 40)x_{i2} + \beta_3x_{i3} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

with x_1 quantitative and $x_{i2} = x_{i3} = \mathbf{1}\{x_{i1} > 40\}$. State and carry out a test of whether the mean response jumps at $x_1 = 40$.

Solution.

Step 1. (The mean function.) For $x_1 \le 40$, $E(y) = \beta_0 + \beta_1x_1$; for $x_1 > 40$,

$$E(y) = (\beta_0 - 40\beta_2 + \beta_3) + (\beta_1 + \beta_2)x_1.$$

So the model consists of two lines with arbitrary intercepts and slopes on either side of 40; β_2 is the change in slope.

Step 2. (The jump.) The left line at $x_1 = 40$ gives $\beta_0 + 40\beta_1$, and the right line tends to $\beta_0 - 40\beta_2 + \beta_3 + 40(\beta_1 + \beta_2) = \beta_0 + 40\beta_1 + \beta_3$ as $x_1 \downarrow 40$. The jump is β_3, so the hypotheses are

$$H_0 : \beta_3 = 0 \quad \text{(continuous at 40)} \qquad \text{vs} \qquad H_1 : \beta_3 \neq 0.$$

Step 3. (The test.) Assume the design $\mathbf{1}$, $\mathbf{x}_1$, $(\mathbf{x}_1 - 40)\mathbf{x}_2$, $\mathbf{x}_3$ has full rank (e.g. two distinct x_1 values on each side of 40; a reparametrization of two lines). Fit and compute

$$t_0 = \frac{\hat{\beta}_3}{\sqrt{c_{33} MSE}}, \qquad MSE = \frac{SSE}{n-4},$$

c_{33} the $(X'X)^{-1}$ entry for β_3. Under H_0, $t_0 \sim t(n-4)$ (Section 6.5); reject when $|t_0| > t_{\alpha/2}(n-4)$. Equivalently compare with the continuous broken line (Section 13.3) by the partial $F_0 = t_0^2$ on $(1, n-4)$ d.f. (Section 8.2). ■

Exercise 13.2 (An indicator alone is the two-sample t test)

Let $y_i = \beta_0 + \beta_1 d_i + \epsilon_i$, $d_i \in \{0, 1\}$, with $n_0, n_1 \geq 1$, $n \geq 3$, and group means $\bar{y}_0, \bar{y}_1$. Show that $\hat{\beta}_0 = \bar{y}_0$, $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$, $MSE = s_p^2$, and the t statistic for $\beta_1 = 0$ is the pooled two-sample t.

Solution [Supplement].

Step 1. This is a simple regression on $x = d$ with the two design points $a = 0$, $b = 1$. By Exercise 3.2, $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$ and the fitted line passes through $(0, \bar{y}_0)$, so $\hat{\beta}_0 = \bar{y}_0$; also $S_{xx} = n_0 n_1 / n$.

Step 2. The fitted value of each observation is its group mean, so

$$SSE = \sum_{d_i=0} (y_i - \bar{y}_0)^2 + \sum_{d_i=1} (y_i - \bar{y}_1)^2, \qquad MSE = \frac{SSE}{n-2} = s_p^2.$$

Step 3. By Section 4.1,

$$t_0 = \frac{\hat{\beta}_1}{\sqrt{MSE / S_{xx}}} = \frac{\bar{y}_1 - \bar{y}_0}{s_p \sqrt{n/(n_0 n_1)}} = \frac{\bar{y}_1 - \bar{y}_0}{s_p \sqrt{1/n_0 + 1/n_1}},$$

which is the pooled two-sample t statistic, with the same $n - 2$ degrees of freedom. ■

Exercise 13.3 (The common slope of parallel lines)

In the parallel-lines model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$ of Section 13.1, let group $g \in \{0, 1\}$ (with $x_2 = g$) have mean $\bar{x}_g$ of x_1, mean $\bar{y}_g$, and within-group sums $S_{xx,g} = \sum_{i \in g} (x_{i1} - \bar{x}_g)^2$, $S_{xy,g} = \sum_{i \in g} (x_{i1} - \bar{x}_g)(y_i - \bar{y}_g)$, with $S_{xx,0} + S_{xx,1} > 0$. Show that

$$\hat{\beta}_1 = \frac{S_{xy,0} + S_{xy,1}}{S_{xx,0} + S_{xx,1}}, \qquad \hat{\beta}_2 = (\bar{y}_1 - \bar{y}_0) - \hat{\beta}_1 (\bar{x}_1 - \bar{x}_0).$$

Solution [Supplement].

Step 1. Theorem 6.5 with $X_1 = (\mathbf{1}, \mathbf{x}_2)$, $X_2 = \mathbf{x}_1$: $\mathrm{col}(X_1)$ is spanned by the orthogonal group indicators $\mathbf{1}_{G_0} = \mathbf{1} - \mathbf{x}_2$, $\mathbf{1}_{G_1} = \mathbf{x}_2$, so $H_1 \mathbf{v}$ replaces entries by group means and $(I - H_1)\mathbf{v}$ subtracts them.

Step 2. Hence $\mathbf{e} = (I - H_1)\mathbf{y}$ has entries $y_i - \bar{y}_{g(i)}$ and $X_{2\cdot1} = (I - H_1)\mathbf{x}_1$ has entries $x_{i1} - \bar{x}_{g(i)}$. Since $\|X_{2\cdot1}\|^2 = S_{xx,0} + S_{xx,1} > 0$, the design has full rank, and

$$\hat{\beta}_1 = \frac{X_{2\cdot1}'\mathbf{e}}{X_{2\cdot1}'X_{2\cdot1}} = \frac{S_{xy,0} + S_{xy,1}}{S_{xx,0} + S_{xx,1}}.$$

Step 3. Residuals are orthogonal to $\mathbf{1}_{G_0}, \mathbf{1}_{G_1}$, so averaging the fit over each group gives $\bar{y}_0 = \hat{\beta}_0 + \hat{\beta}_1\bar{x}_0$ and $\bar{y}_1 = \hat{\beta}_0 + \hat{\beta}_2 + \hat{\beta}_1\bar{x}_1$; subtract.

So the common slope pools within-group information, and $\hat{\beta}_2$ compares group means *adjusted* to a common x_1 (analysis of covariance). ■

Exercise 13.4 (The interaction model is two separate fits)

Fit $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$ ($x_2 \in \{0, 1\}$) to the pooled data, and lines $b_{0g} + b_{1g}x_1$ per group (each with two distinct x_1). Show equal fitted values, $SSE = SSE_0 + SSE_1$, and $\hat{\beta}_0 = b_{00}$, $\hat{\beta}_1 = b_{10}$, $\hat{\beta}_2 = b_{01} - b_{00}$, $\hat{\beta}_3 = b_{11} - b_{10}$.

Solution [Supplement].

Step 1. Let Z have columns $\mathbf{1}_{G_0}$, $\mathbf{x}_1\mathbf{1}_{G_0}$, $\mathbf{1}_{G_1}$, $\mathbf{x}_1\mathbf{1}_{G_1}$ (products taken entrywise), the design of the separate fits. The interaction design has columns

$$\mathbf{1} = \mathbf{1}_{G_0} + \mathbf{1}_{G_1}, \quad \mathbf{x}_1 = \mathbf{x}_1\mathbf{1}_{G_0} + \mathbf{x}_1\mathbf{1}_{G_1}, \quad \mathbf{x}_2 = \mathbf{1}_{G_1}, \quad \mathbf{x}_1\mathbf{x}_2 = \mathbf{x}_1\mathbf{1}_{G_1},$$

so $X = ZA$ for a 4×4 matrix A; it is nonsingular, since conversely $\mathbf{1}_{G_0} = \mathbf{1} - \mathbf{x}_2$ and $\mathbf{x}_1\mathbf{1}_{G_0} = \mathbf{x}_1 - \mathbf{x}_1\mathbf{x}_2$ express Z through X.

Step 2. Ordered by group, Z is block diagonal, so $\|\mathbf{y} - Z\mathbf{b}\|^2 = \sum_g \|\mathbf{y}_g - b_{0g}\mathbf{1} - b_{1g}\mathbf{x}_{1,g}\|^2$ is minimized blockwise: the separate lines, $SSE = SSE_0 + SSE_1$. Each block has full rank, so Z and X do.

Step 3. By Exercise 6.3, X and Z give the same fitted values and SSE, so $X\hat{\beta} = Z\hat{\mathbf{b}}$. By Step 1,

$$X\hat{\beta} = \hat{\beta}_0\mathbf{1}_{G_0} + \hat{\beta}_1\mathbf{x}_1\mathbf{1}_{G_0} + (\hat{\beta}_0 + \hat{\beta}_2)\mathbf{1}_{G_1} + (\hat{\beta}_1 + \hat{\beta}_3)\mathbf{x}_1\mathbf{1}_{G_1},$$

and since the columns of Z are linearly independent, $b_{00} = \hat{\beta}_0$, $b_{10} = \hat{\beta}_1$, $b_{01} = \hat{\beta}_0 + \hat{\beta}_2$ and $b_{11} = \hat{\beta}_1 + \hat{\beta}_3$. This justifies the remark after the interaction model in Section 13.1. ■

Chapter 15. Multicollinearity and Biased Estimation

$\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is BLUE with covariance $\sigma^2(X'X)^{-1}$, huge when $X'X$ is nearly singular; a slightly biased, much less variable estimator can then be far more accurate.

We standardize: y centred, regressors centred and scaled ($\sum_i x_{ij} = 0$, $\sum_i x_{ij}^2 = 1$), so X is $n \times p$, $X'X$ is the regressor correlation matrix and $X'\mathbf{y}$ is proportional to their correlations with y. Original-scale coefficients: $\tilde\beta_j = \hat\beta_j\, s_y/s_j$, $\tilde\beta_0 = \bar y - \sum_j \tilde\beta_j \bar x_j$.

15.1 Multicollinearity

Multicollinearity means that the regressors are (nearly) linearly dependent. Exact dependence makes X rank-deficient and $\hat\beta$ undefined; near dependence makes $(X'X)^{-1}$ huge and $\hat\beta$ unstable.

Example 15.1 (Two regressors)

With two standardized regressors with correlation r_{12},

$$X'X = \begin{pmatrix} 1 & r_{12} \\ r_{12} & 1 \end{pmatrix}, \qquad \operatorname{Var}\hat\beta = \frac{\sigma^2}{1-r_{12}^2}\begin{pmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{pmatrix},$$

so $\operatorname{Var}\hat\beta_j = \sigma^2/(1-r_{12}^2) \to \infty$ as $|r_{12}| \to 1$, and $\operatorname{Corr}(\hat\beta_1, \hat\beta_2) = -r_{12}$. With $r_{12} = 0.99$ the variances are 50 times those under orthogonal regressors, and confidence intervals $\hat\beta_j \pm t_{\alpha/2}(n-3)\sqrt{MSE/(1-r_{12}^2)}$ are 7 times as wide.

Eigenvalues. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $X'X$,

$$\sum_j \operatorname{Var}\hat\beta_j = \sigma^2 \operatorname{tr}(X'X)^{-1} = \sigma^2 \sum_j \frac{1}{\lambda_j},$$

so a single small eigenvalue blows up the total variance. The eigenvector of a small eigenvalue shows which combination of regressors is nearly constant.

Theorem 15.1 (Variance inflation factor)

Let R_j^2 be the coefficient of determination of the regression of x_j on the other regressors. Then c_{jj}, the j-th diagonal entry of $(X'X)^{-1}$, equals $(1-R_j^2)^{-1}$, so

$$\operatorname{Var}\hat\beta_j = \frac{\sigma^2}{1-R_j^2}.$$

Proof. Write $X = (\mathbf{x}_j, X_{-j})$. By the partitioned-inverse formula,

$$c_{jj} = \{\mathbf{x}_j'\mathbf{x}_j - \mathbf{x}_j'X_{-j}(X_{-j}'X_{-j})^{-1}X_{-j}'\mathbf{x}_j\}^{-1} = \{\mathbf{x}_j'(I-H_{-j})\mathbf{x}_j\}^{-1},$$

the inverse residual sum of squares of $\mathbf{x}_j$ on X_{-j}, which is $1-R_j^2$ since $\mathbf{x}_j'\mathbf{x}_j = 1$. ∎

$\mathrm{VIF}_j = (1-R_j^2)^{-1}$ is the **variance inflation factor**: the factor by which $\operatorname{Var}\hat\beta_j$ exceeds its value under orthogonal regressors. $\mathrm{VIF}_j > 10$ (i.e. $R_j^2 > 0.9$) is a common warning sign.

15.2 Ridge regression

Motivation. Ridge adds kI to $X^\top X$: biased, but for suitable k smaller MSE than least squares.

Measure the accuracy of an estimator b by the total mean squared error $E\|b-\beta\|^2$. For least squares,

$$E\|\hat\beta-\beta\|^2 = \operatorname{tr}\operatorname{Var}\hat\beta = \sigma^2\sum_j \frac{1}{\lambda_j}.$$

Definition (Ridge estimator)

For $k>0$, $\hat\beta(k)=(X'X+kI_p)^{-1}X'\mathbf{y}$, the minimizer of $\|\mathbf{y}-X\beta\|^2+k\|\beta\|^2$.

Adding k lifts all eigenvalues, conditioning $X'X+kI$. If $X'X=I$, $\hat\beta(k)=\hat\beta/(1+k)$: ridge **shrinks** towards $\mathbf{0}$. In general $\hat\beta(k)=(X'X+kI)^{-1}X'X\hat\beta$, with $E\hat\beta(k)=(X'X+kI)^{-1}X'X\beta\neq\beta$.

Canonical form. Let $X'X=P\Lambda P'$ with P orthogonal and $\Lambda=\operatorname{diag}(\lambda_1,\dots,\lambda_p)$. With $Z=XP$ and $\alpha=P'\beta$, the model is $\mathbf{y}=Z\alpha+\epsilon$ with $Z'Z=\Lambda$. Then $\hat\alpha=\Lambda^{-1}Z'\mathbf{y}$, $\operatorname{Var}\hat\alpha=\sigma^2\Lambda^{-1}$, and

$$\hat\alpha(k)=P'\hat\beta(k)=(\Lambda+kI)^{-1}Z'\mathbf{y}, \qquad \hat\alpha_j(k)=\frac{\lambda_j}{\lambda_j+k}\hat\alpha_j :$$

each canonical coefficient is shrunk, most strongly in directions with small λ_j. Since P is orthogonal, $\|\hat\beta(k)-\beta\|=\|\hat\alpha(k)-\alpha\|$.

Theorem 15.2 (Mean squared error of ridge)

$$E\|\hat\beta(k)-\beta\|^2=\sigma^2\sum_{j=1}^{p}\frac{\lambda_j}{(\lambda_j+k)^2}+k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{(\lambda_j+k)^2}.$$

The first term (total variance) decreases in k and the second (squared bias) increases. There is always a $k>0$ with $E\|\hat\beta(k)-\beta\|^2<E\|\hat\beta-\beta\|^2$.

Proof [Supplement]. $\hat\alpha_j(k)$ has mean $\frac{\lambda_j}{\lambda_j+k}\alpha_j$ and variance $\frac{\lambda_j^2}{(\lambda_j+k)^2}\cdot\frac{\sigma^2}{\lambda_j}$; its bias is $-\frac{k}{\lambda_j+k}\alpha_j$. Adding variance and squared bias over j gives the formula. Call it $g(k)$. Then $g(0)=\sigma^2\sum 1/\lambda_j=E\|\hat\beta-\beta\|^2$, and

$$g'(k)=-2\sigma^2\sum_j\frac{\lambda_j}{(\lambda_j+k)^3}+2k\sum_j\frac{\lambda_j\alpha_j^2}{(\lambda_j+k)^3}.$$

At $k=0$, $g'(0)=-2\sigma^2\sum_j\lambda_j^{-2}<0$, so $g(k)<g(0)$ for small $k>0$. ∎

In the orthonormal case $X'X=I$, $g(k)=\frac{p\sigma^2+k^2\|\alpha\|^2}{(1+k)^2}$, minimized at $k=p\sigma^2/\|\beta\|^2$ (set $g'(k)=0$). More generally, a separate constant k_j for each canonical direction, $\hat\alpha_j=\frac{\lambda_j}{\lambda_j+k_j}\hat\alpha_j$ (generalized ridge), has mean squared error $\sum_j\frac{\sigma^2\lambda_j+k_j^2\alpha_j^2}{(\lambda_j+k_j)^2}$, minimized at $k_j=\sigma^2/\alpha_j^2$.

Choosing k. The optimal k depends on unknown σ^2 and β. Practical choices:

- the **ridge trace**: plot $\hat\beta_j(k)$ against k and take the smallest k at which the coefficients stabilize;
- the plug-in rule $\hat k=p\hat\sigma^2/\hat\beta'\hat\beta$ (Hoerl, Kennard and Baldwin), possibly iterated with $\hat\beta(k)$ in place of $\hat\beta$;

- cross-validation, e.g. minimizing $\text{PRESS}(k) = \sum_i \{y_i - \mathbf{x}_i' \hat{\beta}_{(i)}(k)\}^2$. As in Theorem 10.3, with $A(k) = X(X'X + kI)^{-1}X' = (a_{ij})$ and $e_i = y_i - \mathbf{x}_i'\hat{\beta}(k)$,

$$\text{PRESS}(k) = \sum_{i=1}^{n} \Big(\frac{e_i}{1 - a_{ii}} \Big)^2,$$

so no refitting is needed (leave-one-out cross-validation).

15.3 Principal component regression

Let $\mathbf{e}_1, \ldots, \mathbf{e}_p$ be orthonormal eigenvectors of $X'X$ with eigenvalues $l_1 \geq \cdots \geq l_p > 0$, and $E_r = (\mathbf{e}_1, \ldots, \mathbf{e}_r)$. The **principal components** are the columns of $Z_r = XE_r$, and they are orthogonal: $Z_r'Z_r = \text{diag}(l_1, \ldots, l_r)$.

All components ($r = p$). $X\beta = XE_pE_p'\beta = Z\alpha$ with $\alpha = E_p'\beta$, and

$$\hat{\beta} = E_p\hat{\alpha} = E_p(Z'Z)^{-1}Z'\mathbf{y} = \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}, \qquad \text{Var}\,\hat{\beta} = \sigma^2 \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

which is just least squares.

Leading components ($r < p$). Regress $\mathbf{y}$ on Z_r only and transform back:

$$\tilde{\beta}_r = E_r(Z_r'Z_r)^{-1}Z_r'\mathbf{y} = \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}.$$

Theorem 15.3 (Principal component regression)

$$E\tilde{\beta}_r = \beta - \sum_{k=r+1}^{p} \mathbf{e}_k\mathbf{e}_k'\beta, \qquad \text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

and

$$E\|\tilde{\beta}_r - \beta\|^2 = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k} + \sum_{k=r+1}^{p} (\mathbf{e}_k'\beta)^2.$$

Proof. Using $X'X = \sum_h l_h\mathbf{e}_h\mathbf{e}_h'$ and orthonormality, $E\tilde{\beta}_r = \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'X\beta = \sum_{k\leq r} \mathbf{e}_k\mathbf{e}_k'\beta$, and $\beta = \sum_{k\leq p} \mathbf{e}_k\mathbf{e}_k'\beta$. Next,

$$\text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k,k'\leq r} \frac{1}{l_k l_{k'}}\mathbf{e}_k\mathbf{e}_k'X'X\mathbf{e}_{k'}\mathbf{e}_{k'}' = \sigma^2 \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'.$$

The trace of the variance plus the squared length of the bias gives the mean squared error. ■

Dropping the smallest-l_k components removes the largest variances σ^2/l_k at bias $\sum_{k>r}(\mathbf{e}_k'\beta)^2$, worthwhile when those directions carry little of β. PCR is "hard" ridge: ridge shrinks each canonical coefficient by $l_k/(l_k + k)$, PCR keeps r and zeros the rest.

Choosing r. Keep components until $\sum_{k\leq r} l_k / \sum_k l_k$ is large, look for a scree-plot elbow, or cross-validate. Components ignore y; partial least squares builds components correlated with y.

15.4 Selected exercises

Motivation. The VIF on the original scale, ridge as least squares on augmented data, and ridge as norm-constrained least squares.

Exercise 15.1 (The VIF on the original scale)

In $Y_i = \beta_0+\beta_1 x_{i1}+\cdots+\beta_p x_{ip}+\epsilon_i$ (uncorrelated errors, variance σ^2, unstandardized regressors), let $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2 > 0$ and R_j^2 the R^2 of x_j on the other regressors (with intercept). Show that

$$\operatorname{Var}\hat{\beta}_j = \frac{\sigma^2}{S_{jj}}\,\mathrm{VIF}_j, \qquad \mathrm{VIF}_j = (1 - R_j^2)^{-1}.$$

Solution.

Step 1. (Centring.) Let $C = I - J_n/n$, the projection onto $\mathbf{1}^\perp$, and $\tilde{X} = CX_s$, where X_s holds the p regressor columns. By Theorem 6.5 with $X_1 = \mathbf{1}$, the slope vector is $\hat{\beta}_s = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'C\mathbf{y} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\mathbf{y}$, so $\operatorname{Var}\hat{\beta}_s = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'(\sigma^2 I)\tilde{X}(\tilde{X}'\tilde{X})^{-1} = \sigma^2(\tilde{X}'\tilde{X})^{-1}$.

Step 2. (Partitioned inverse.) Write $\tilde{X} = (\tilde{\mathbf{x}}_j, \tilde{X}_{-j})$ and $\tilde{H}_{-j} = \tilde{X}_{-j}(\tilde{X}'_{-j}\tilde{X}_{-j})^{-1}\tilde{X}'_{-j}$. As in the proof of Theorem 15.1, the j-th diagonal entry of $(\tilde{X}'\tilde{X})^{-1}$ is $\{\tilde{\mathbf{x}}_j'(I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j\}^{-1}$.

Step 3. (A residual sum of squares.) $(\mathbf{1}, X_{-j})$ has projection $J_n/n + \tilde{H}_{-j}$ (centred $\tilde{X}_{-j} \perp \mathbf{1}$; cf. Theorem 10.1(d)), and $\tilde{H}_{-j}\mathbf{1} = \mathbf{0}$, so the residual of $\mathbf{x}_j$ is

$$(I - J_n/n - \tilde{H}_{-j})\mathbf{x}_j = \tilde{\mathbf{x}}_j - \tilde{H}_{-j}\tilde{\mathbf{x}}_j = (I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j,$$

whose squared length is $\tilde{\mathbf{x}}_j'(I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j$. The total sum of squares of x_j is S_{jj}, so this residual sum of squares equals $S_{jj}(1 - R_j^2)$.

Step 4. So $\operatorname{Var}\hat{\beta}_j = \sigma^2/\{S_{jj}(1 - R_j^2)\}$: σ^2/S_{jj} is the simple-regression variance (Theorem 4.1(a)), attained when x_j is uncorrelated with the others, and VIF_j the collinearity inflation; $S_{jj} = 1$ gives Theorem 15.1. ∎

Exercise 15.2 (Ridge regression as augmented least squares)

Let $E(\mathbf{y} \mid X) = X\beta$ and $\operatorname{Var}(\mathbf{y} \mid X) = \sigma^2 I_n$, X of size $n \times p$ with singular value decomposition $X = UDV'$ (U of size $n \times p$ with orthonormal columns, $D = \operatorname{diag}(d_1, \ldots, d_p)$ with $d_j \geq 0$, V orthogonal). For $\lambda > 0$ let

$$X_* = \begin{pmatrix} X \\ \sqrt{\lambda} I_p \end{pmatrix}, \qquad \mathbf{y}_* = \begin{pmatrix} \mathbf{y} \\ \mathbf{0}_p \end{pmatrix}, \qquad \hat{\beta} = (X_*'X_*)^{-1}X_*'\mathbf{y}_*.$$

Express $E(\hat{\beta} \mid X)$ and $\operatorname{Var}(\hat{\beta} \mid X)$ through U, D, V, β and σ.

Solution.

Step 1. $X_*'X_* = X'X + \lambda I$ and $X_*'\mathbf{y}_* = X'\mathbf{y}$, so $\hat{\beta} = (X'X + \lambda I)^{-1}X'\mathbf{y}$ is the ridge estimator with $k = \lambda$; it is defined even if X is rank deficient, because $X'X + \lambda I$ is positive definite.

Step 2. $X'X = VDU'UDV' = VD^2V'$ and $X'X + \lambda I = V(D^2 + \lambda I)V'$, so $(X'X + \lambda I)^{-1} = V(D^2 + \lambda I)^{-1}V'$ and

$$\hat{\beta} = V(D^2 + \lambda I)^{-1}V'VDU'\mathbf{y} = V(D^2 + \lambda I)^{-1}DU'\mathbf{y}.$$

Step 3. (Mean.) $E(\hat{\beta} \mid X) = V(D^2 + \lambda I)^{-1}DU'UDV'\beta = V\operatorname{diag}\Big(\frac{d_j^2}{d_j^2 + \lambda}\Big)V'\beta.$

Step 4. (Variance.) With $M = V(D^2+\lambda I)^{-1}DU'$, $\operatorname{Var}(\hat\beta \mid X) = \sigma^2 MM'$ and $U'U = I$ give

$$\operatorname{Var}(\hat\beta \mid X) = \sigma^2 V(D^2+\lambda I)^{-1}D^2(D^2+\lambda I)^{-1}V' = \sigma^2 V \operatorname{diag}\Big(\frac{d_j^2}{(d_j^2+\lambda)^2}\Big)V'.$$

In coordinates $\alpha = V'\beta$ these are the canonical formulas (Section 15.2) with $\lambda_j = d_j^2$: shrinkage by $d_j^2/(d_j^2+\lambda)$. The p extra rows are pseudo-observations $0 = \sqrt{\lambda}\beta_j +$ error pulling β towards $\mathbf{0}$. ■

Exercise 15.3 (Ridge as constrained least squares)

Let X $(n\times p)$ have full column rank, $\hat\beta$ be the least squares estimator, $Q(\beta) = (\beta-\hat\beta)'X'X(\beta-\hat\beta)$, and $c > 0$.
(a) Find the minimizer of $Q(\beta)$ subject to $\beta'\beta \le c$.
(b) For fixed $k \ge 0$, find the mean and variance of $\hat\beta(k) = (X'X+kI)^{-1}X'\mathbf{y}$.
(c) Using the singular value decomposition $X = UDV'$, show that $\hat\beta(k)$ shrinks $\hat\beta$ towards $\mathbf{0}$.

Solution.

(a) If $\hat\beta'\hat\beta \le c$, then $\hat\beta$ is feasible with $Q(\hat\beta) = 0$, and it is the unique minimizer since $X'X$ is positive definite. Assume now $\hat\beta'\hat\beta > c$.

Step 1. For $k \ge 0$, $Q(\beta)+k\|\beta\|^2$ is strictly convex with gradient $2X'X(\beta-\hat\beta)+2k\beta$, which vanishes at $(X'X+kI)^{-1}X'X\hat\beta = (X'X+kI)^{-1}X'\mathbf{y} = \hat\beta(k)$. So $\hat\beta(k)$ is its unique minimizer.

Step 2. With $X'X = P\Lambda P'$ and $\hat\alpha = P'\hat\beta$, $\|\hat\beta(k)\|^2 = \sum_j \lambda_j^2\hat\alpha_j^2/(\lambda_j+k)^2$. This is continuous and strictly decreasing in k (since $\hat\alpha \ne \mathbf{0}$), equals $\|\hat\beta\|^2 > c$ at $k = 0$, and tends to 0. Hence there is a unique $k^* > 0$ with $\|\hat\beta(k^*)\|^2 = c$.

Step 3. For any feasible $\beta \ne \hat\beta(k^*)$, by Step 1,

$$Q(\beta) > Q(\hat\beta(k^*)) + k^*(\|\hat\beta(k^*)\|^2 - \|\beta\|^2) = Q(\hat\beta(k^*)) + k^*(c - \|\beta\|^2) \ge Q(\hat\beta(k^*)).$$

So the constrained minimizer is the ridge estimator $\hat\beta(k^*)$, the ridge constant being the Lagrange multiplier of the constraint.

(b) $\hat\beta(k) = M\mathbf{y}$ with $M = (X'X+kI)^{-1}X'$, so

$$E\hat\beta(k) = (X'X+kI)^{-1}X'X\beta, \qquad \operatorname{Var}\hat\beta(k) = \sigma^2(X'X+kI)^{-1}X'X(X'X+kI)^{-1}.$$

These formulas are for a fixed k; the k^* of (a) depends on $\mathbf{y}$, so they describe the constrained estimator only conditionally on its multiplier.

(c) As in Exercise 15.2, $\hat\beta(k) = V\operatorname{diag}\{d_j^2/(d_j^2+k)\}V'\hat\beta$, $d_j > 0$; each coordinate of $V'\hat\beta$ is multiplied by a factor in $(0,1)$ for $k > 0$, so $\|\hat\beta(k)\| < \|\hat\beta\|$ $(\hat\beta \ne \mathbf{0})$, with the smallest singular values shrunk most. ■

Chapter 16. Logistic Regression

16.1 A model for a binary response

For $y \in \{0,1\}$, $E(y \mid x) = P(y=1 \mid x) =: \pi(x)$. The linear model fails: fitted values leave $[0,1]$, errors are two-valued, and $\mathrm{Var}(y \mid x) = \pi(x)\{1-\pi(x)\}$ varies. $\pi(x)$ is often S-shaped, so we model

$$\pi(x) = G(\beta_0 + \beta_1 x)$$

with a cdf G (the inverse **link**): logistic $G(t) = e^t/(1+e^t)$, probit Φ, or Gumbel $\exp(-e^{-t})$. The logistic gives the **logistic regression model**

$$\pi(x) = \frac{\exp(\beta_0+\beta_1 x)}{1+\exp(\beta_0+\beta_1 x)}, \qquad \text{equivalently} \qquad \operatorname{logit}\pi(x) = \log\frac{\pi(x)}{1-\pi(x)} = \beta_0 + \beta_1 x.$$

The curve increases in x if $\beta_1 > 0$ and decreases if $\beta_1 < 0$.

Odds and odds ratios. The **odds** of $y=1$ are $\pi/(1-\pi) = P(y=1)/P(y=0) \in [0,\infty)$, with $\pi = \text{odds}/(1+\text{odds})$. Under the logistic model

$$\frac{\pi(x)}{1-\pi(x)} = e^{\beta_0}(e^{\beta_1})^x, \qquad \frac{\pi(x+1)/\{1-\pi(x+1)\}}{\pi(x)/\{1-\pi(x)\}} = e^{\beta_1}.$$

So e^{β_1} is the **odds ratio** per unit of x, the same at every x; for binary x it compares the groups, e.g. $\hat\beta_1 = 2.0$ means $e^2 \approx 7.4$ times higher odds when exposed.

16.2 Estimation

Motivation. Logistic regression models the log-odds linearly, fitted by maximum likelihood via IRLS.

With $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ip})'$ and $\pi_i = \pi(\mathbf{x}_i) = e^{\mathbf{x}_i'\beta}/(1+e^{\mathbf{x}_i'\beta})$, independent observations $y_i \sim \text{Bernoulli}(\pi_i)$ give the likelihood $L(\beta) = \prod \pi_i^{y_i}(1-\pi_i)^{1-y_i}$ and

$$l(\beta) = \sum_{i=1}^n \{y_i \log\pi_i + (1-y_i)\log(1-\pi_i)\} = \sum_{i=1}^n \{y_i \mathbf{x}_i'\beta - \log(1+e^{\mathbf{x}_i'\beta})\}.$$

Theorem 16.1

With $\pi(\beta) = (\pi_1,\ldots,\pi_n)'$ and $W(\beta) = \mathrm{diag}\{\pi_i(1-\pi_i)\}$,

$$\frac{\partial l}{\partial \beta} = X'\{\mathbf{y} - \pi(\beta)\}, \qquad \frac{\partial^2 l}{\partial\beta\partial\beta'} = -X'W(\beta)X.$$

The Hessian is negative definite (full column rank), so the MLE, if it exists, is the unique root of $X'\{\mathbf{y}-\pi(\hat\beta)\} = \mathbf{0}$.

Proof. $\frac{d}{dt}\log(1+e^t) = \frac{e^t}{1+e^t}$ and $\frac{d^2}{dt^2}\log(1+e^t) = \frac{e^t}{(1+e^t)^2} = G(t)\{1-G(t)\}$; apply the chain rule with $t_i = \mathbf{x}_i'\beta$. Since $0<\pi_i<1$, $\mathbf{a}'X'WX\mathbf{a} = \sum_i \pi_i(1-\pi_i)(\mathbf{x}_i'\mathbf{a})^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$. ■

The likelihood equations are nonlinear and are solved by Newton–Raphson:

$$\beta^{(t+1)} = \beta^{(t)} + \{X'W^{(t)}X\}^{-1}X'\{\mathbf{y}-\pi^{(t)}\} = \{X'W^{(t)}X\}^{-1}X'W^{(t)}\mathbf{z}^{(t)},$$

with the **working response**

$$\mathbf{z}^{(t)} = X\beta^{(t)} + \{W^{(t)}\}^{-1}\{\mathbf{y} - \pi^{(t)}\}.$$

Each step is a weighted least squares regression of $\mathbf{z}^{(t)}$ on X with weights $\pi_i^{(t)}(1-\pi_i^{(t)})$:

$$\beta^{(t+1)} = \arg\min_{\beta}\{\mathbf{z}^{(t)} - X\beta\}'W^{(t)}\{\mathbf{z}^{(t)} - X\beta\},$$

hence **iteratively reweighted least squares** (IRLS), usually from $\beta^{(0)} = \mathbf{0}$. The Hessian is free of $\mathbf{y}$, so Newton–Raphson is Fisher scoring, and $\{X'W(\hat\beta)X\}^{-1}$ estimates $\operatorname{Cov}\hat\beta$. Wald: $\hat\beta_j/\operatorname{se}(\hat\beta_j) \approx N(0,1)$; LR: $2\{l(\hat\beta_F) - l(\hat\beta_R)\}$ vs chi-square.

If the two classes can be perfectly separated by a hyperplane in x, l increases without bound along a direction and the MLE does not exist.

16.3 Two-by-two tables

Motivation. With one binary predictor the model is a 2×2 table and the slope a log odds ratio.

Binary exposure (x) and outcome (y) data arise from three designs:

- **cohort** (prospective) studies follow exposed and unexposed groups and record outcomes;
- **case–control** (retrospective) studies sample cases ($y = 1$) and controls ($y = 0$) and record past exposure;
- **cross-sectional** studies sample individuals and record both at one time.

With counts n_{xy}:

	$y=0$	$y=1$	total
$x=0$	n_{00}	n_{01}	n_{0+}
$x=1$	n_{10}	n_{11}	n_{1+}

The **odds ratio** is $OR = \dfrac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)}$ with $\pi_x = P(y=1 \mid x)$, estimated by $\widehat{OR} = \dfrac{n_{11}n_{00}}{n_{10}n_{01}}$.

Theorem 16.2

In the logistic model $\operatorname{logit} P(y=1 \mid x) = \beta_0 + \beta_1 x$ with binary x, the MLEs satisfy

$$\hat\pi_1 = \frac{e^{\hat\beta_0+\hat\beta_1}}{1+e^{\hat\beta_0+\hat\beta_1}} = \frac{n_{11}}{n_{1+}}, \qquad \hat\pi_0 = \frac{e^{\hat\beta_0}}{1+e^{\hat\beta_0}} = \frac{n_{01}}{n_{0+}}, \qquad e^{\hat\beta_1} = \frac{n_{11}n_{00}}{n_{10}n_{01}}.$$

Proof. The log-likelihood is

$$l = n_{11}(\beta_0+\beta_1) + n_{01}\beta_0 - n_{1+}\log(1+e^{\beta_0+\beta_1}) - n_{0+}\log(1+e^{\beta_0}).$$

The model is saturated (two parameters for two probabilities), so by invariance the MLEs of π_1, π_0 are the sample proportions n_{11}/n_{1+} and n_{01}/n_{0+}. Then $e^{\hat\beta_1} = \frac{\hat\pi_1/(1-\hat\pi_1)}{\hat\pi_0/(1-\hat\pi_0)} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ∎

Why the odds ratio. In case–control studies row proportions do not estimate π_x (case and control numbers are fixed), but the odds ratio is symmetric in x and y:

$$\frac{P(y=1 \mid x=1)P(y=0 \mid x=0)}{P(y=0 \mid x=1)P(y=1 \mid x=0)} = \frac{P(x=1 \mid y=1)P(x=0 \mid y=0)}{P(x=0 \mid y=1)P(x=1 \mid y=0)},$$

both being $\frac{p_{11}p_{00}}{p_{10}p_{01}}$; so β_1 is estimable from case–control data, only the intercept being distorted.

Example 16.1

140 subjects get a supplement, 140 a placebo; 17 and 31 catch a cold. With $x = 1$ for the supplement, $\widehat{OR} = \frac{17 \cdot 109}{123 \cdot 31} = 0.49$, $\hat{\beta}_1 = -0.72$, SE $\sqrt{\frac{1}{17} + \frac{1}{123} + \frac{1}{31} + \frac{1}{109}} = 0.33$, Wald -2.19: significant at 5%.

16.4 Selected exercises

Motivation. Fitted and observed counts agree, and the SE of Example 16.1 comes from the information matrix.

Exercise 16.1 (Fitted counts equal observed counts)

(a) Show that the MLE of a logistic regression satisfies $X'\hat{\pi} = X'\mathbf{y}$; in particular, with an intercept, $\sum_i \hat{\pi}_i = \sum_i y_i$.
(b) For a binary x with all four cell counts n_{xy} positive, derive Theorem 16.2 directly from (a).

Solution [Supplement].

(a) The MLE solves $X'\{\mathbf{y} - \pi(\hat{\beta})\} = \mathbf{0}$ (Theorem 16.1); the intercept row gives $\sum \hat{\pi}_i = \sum y_i$, and each regressor $\sum_i x_{ij}\hat{\pi}_i = \sum_i x_{ij} y_i$.

(b) With $\mathbf{x}_i = (1, x_i)'$ and $x_i \in \{0, 1\}$, $\hat{\pi}_i$ takes the value $\hat{\pi}_1$ when $x_i = 1$ and $\hat{\pi}_0$ when $x_i = 0$. The two equations of (a) read

$$n_{0+}\hat{\pi}_0 + n_{1+}\hat{\pi}_1 = n_{01} + n_{11}, \qquad n_{1+}\hat{\pi}_1 = n_{11}.$$

So $\hat{\pi}_1 = n_{11}/n_{1+}$, $\hat{\pi}_0 = n_{01}/n_{0+}$ (in $(0, 1)$), and $\hat{\beta}_0 = \operatorname{logit} \hat{\pi}_0$, $\hat{\beta}_1 = \operatorname{logit} \hat{\pi}_1 - \operatorname{logit} \hat{\pi}_0$ are the unique MLE (strict concavity), with $e^{\hat{\beta}_1} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ■

Exercise 16.2 (Standard error of the log odds ratio)

In the setting of Exercise 16.1(b), show that the entry of $\{X'W(\hat{\beta})X\}^{-1}$ for β_1 equals

$$\frac{1}{n_{11}} + \frac{1}{n_{10}} + \frac{1}{n_{01}} + \frac{1}{n_{00}},$$

the squared standard error used in Example 16.1.

Solution [Supplement].

Step 1. $X'WX = \sum_i w_i \mathbf{x}_i \mathbf{x}_i'$ with $w_i = \hat{\pi}_i(1 - \hat{\pi}_i)$. Grouping by x, with $a = n_{0+}\hat{\pi}_0(1 - \hat{\pi}_0)$ and $b = n_{1+}\hat{\pi}_1(1 - \hat{\pi}_1)$,

$$X'WX = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} a + b & b \\ b & b \end{pmatrix}.$$

Step 2. Its determinant is $(a+b)b - b^2 = ab > 0$, and the $(2, 2)$ entry of the inverse is $(a+b)/(ab) = 1/a + 1/b$.

Step 3. By Exercise 16.1(b), $a = n_{0+} \cdot \frac{n_{01}}{n_{0+}} \cdot \frac{n_{00}}{n_{0+}} = \frac{n_{01} n_{00}}{n_{0+}}$, so $\frac{1}{a} = \frac{n_{00} + n_{01}}{n_{01} n_{00}} = \frac{1}{n_{01}} + \frac{1}{n_{00}}$. In the same way $\frac{1}{b} = \frac{1}{n_{11}} + \frac{1}{n_{10}}$. Adding gives the claim. ■